

Coherence Epistemology for Inter-Species Communication: A Black-Box Framework for AI-Mediated Bidirectional Understanding Across Biological and Artificial Systems

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Abstract

Several converging traditions—Dennett’s intentional stance, Skinner’s radical behaviorism, Friston’s free-energy principle—have, from different directions, argued that the study of behavior need not posit privileged access to internal mental states. Yet no unified formal framework exists that (a) integrates multichannel observable data, (b) provides a concrete metric for communicative success, (c) accommodates individual variation within species-level models, and (d) derives non-trivial bounds on achievable performance. This paper synthesizes these traditions into a single operational framework—*coherence epistemology*—and develops its mathematical consequences. We formalize inter-system communication as a black-box prediction problem in which a multichannel AI mediator constructs probabilistic behavioral landscapes from observable streams—neural, kinematic, physiological, acoustic, and contextual—without on-

tological commitment to internal states. The framework is grounded in variational latent-state models and hierarchical Bayesian individuation. Beyond organizing existing constructs, we derive three novel results: (1) an information-theoretic upper bound on achievable coherence given channel bandwidth and noise, (2) a sufficient condition for cross-species semantic alignment based on shared dynamical topology, and (3) a convergence guarantee for the hierarchical individuation procedure under regularity assumptions. We present a detailed experimental protocol for domestic canine subjects and a computational simulation demonstrating the framework on synthetic multimodal data. The paper situates this work within the intellectual lineage of behaviorism, predictive processing, and enactivism, and discusses the substantial ethical challenges of deploying such systems in practice.

Keywords: inter-species communication, black-box epistemology, predictive coherence, zoosemantics, variational inference, brain–computer interface, AI-mediated translation, animal cognition, radical behaviorism, predictive processing

Contents

1	Introduction	3
2	Intellectual Lineage and Literature Review	5
2.1	Radical Behaviorism and Its Underappreciated Sophistication	5
2.2	Predictive Processing and the Free-Energy Principle	6
2.3	Dennett’s Intentional Stance	6
2.4	Neural Decoding and Brain–Computer Interfaces	6
2.5	AI-Assisted Bioacoustics	7
2.6	Latent-Variable Models for Behavioral and Neural Data	7
2.7	The Attitude–Behavior Gap in Human Self-Report	7
3	The Formal Framework	8
3.1	The Black-Box Postulate	8
3.2	The Multichannel Observable Stream	8
3.3	The Predictive Model	9
3.4	Latent Encoding as Computational Convenience	9
4	Novel Formal Results	9
4.1	Information-Theoretic Bound on Achievable Coherence	9
4.2	Conditions for Cross-Species Semantic Alignment	11
4.3	Convergence of Hierarchical Individuation	12
5	Hierarchical Zoosemantic Architecture	13
6	The Coherence Metric	13
7	Experimental Protocol and Simulation	14
7.1	Proposed Protocol: Domestic Canine Subjects	14
7.2	Computational Simulation: Toy Models	15
8	Falsifiable Predictions	17
9	Applications	19
9.1	Animal–Human Communication	19
9.2	Human–Human Communication	19
9.3	AI Transparency	19

10 Ethical Considerations	19
10.1 The Illusion of Understanding	19
10.2 Non-Consenting Subjects	20
10.3 Commodification and Misuse	20
10.4 The Risk of Justifying Harmful Practices	20
11 Broader Theoretical Context	21
12 Self-Critique and Resolution	21
12.1 Critique 1: The Anthropomorphization Risk	21
12.2 Critique 2: The Validation Problem	21
12.3 Critique 3: Scalability Across Neural Architectures	22
12.4 Critique 4: Circularity in Cross-Species Alignment	22
12.5 Critique 5: Relationship to Behaviorism	22
13 Conclusion	22
14 Peer Reviews and Revision History	29
Appendices: Situational Scenarios	34
A Scenario A: Domestic Cat	34
B Scenario B: Domestic Dog	36
C Scenario C: Human–Human (Clinical Setting)	39
D Scenario D: Human–AI (Language Model)	41
E Scenario E: Hypothetical Extraterrestrial Signal	43
F Proof of Theorem 2 (Alignment Sufficiency)	45
G Simulation Code	46

1 Introduction

The history of attempts to communicate with non-human agents is largely a history of asking the wrong question. From early experiments teaching chimpanzees sign language [1] to contemporary efforts decoding cetacean vocalizations [2], the guiding assumption has been that communication requires access to what another being “thinks” or “means.” This assumption, which we shall call the *transparency thesis*, holds that successful communication is a function of the degree to which one system can reconstruct the internal states of another.

The transparency thesis has deep roots. In Western philosophy, the equation of *logos* (rational speech) with *nous* (mind) stretches from Aristotle through Descartes [3]. In cognitive science, the dominant paradigms—from the computational theory of mind to predictive processing—model cognition as an internal process that is, in principle, accessible to sufficiently powerful observation [4,5]. In ethology, the transparency thesis manifests as the goal of determining what an animal “really” feels or intends [6].

This paper argues that the transparency thesis is not merely technically difficult to satisfy but epistemologically incoherent. The argument proceeds in three steps:

1. **Internal states are not observable.** For any system of sufficient complexity, the relationship between internal dynamics and external behavior is many-to-many [7,8].
2. **Even perfect internal-state knowledge does not guarantee behavioral prediction.** Internal states are one variable in a dynamical system that also includes environmental context, physiological state, interaction history, and stochastic fluctuations [9,10]. Knowing the thought does not determine the action.
3. **The same limitation applies to human self-knowledge.** Humans routinely fail to predict their own behavior from their own thoughts [11,12]. Conscious awareness of a “decision” often follows the neural commitment to that decision [13].

These observations converge on a single conclusion: no channel of information—not speech, not neural recording, not introspection—provides reliable access to the interior of any sufficiently complex system.

What, then, is communication? If it cannot be grounded in transparency, it must be grounded in *coherence*: the degree to which one system’s model of another system’s behavior matches the actual behavior over time. This reframing is not new. It draws explicitly on Dennett’s intentional stance [7], which treats mental-state attribution as a predictive strategy; on Skinner’s radical behaviorism [14], which insisted on prediction and control of behavior without appeal to internal states; on Friston’s free-energy principle [5], which formalizes organisms as systems minimizing prediction error over sensory streams; and on the enactivist tradition of Varela, Thompson, and Rosch [15], which locates cognition in the coupling between organism and environment rather than in internal representations.

The vision: giving voice to those who cannot speak

Before proceeding to the formalism, we wish to state plainly what we are trying to build. The goal is an AI system that **speaks for your cat, your dog, your non-verbal child, your aphasic parent**—in first person, in a voice you can customize, with a confidence that tracks how well the system actually predicts their behavior.

The system watches. It listens. It measures heartbeat, posture, gaze, movement, vocalization. From these streams, it builds a predictive model of what the subject will do next. When the model predicts well—when the cat approaches and the system had said “I’m coming to you”—the voice is confident. When the model fails to predict—when something unprecedented happens—the voice says “I don’t know what’s happening to me.” The quality of the voice is the quality of the prediction. Nothing more.

Crucially, **it does not matter what the subject “really thinks.”** We make no claim about phenomenology, consciousness, qualia, or inner experience. The cat may have rich subjective experience or none at all; the framework operates identically in either case. What matters is behavioral coherence: does the model predict what the subject actually does? If yes, the voice is accurate in the only sense that is empirically testable. If no, the voice degrades, and it says so.

The rendering interface—the voice itself—is fully customizable. The human user chooses the timbre, the register, the language, the level of detail. One user may want a warm, childlike voice for their cat; another may want terse clinical reports. One may prefer probabilistic confidence intervals; another may want simple emotional labels. The framework separates the *predictive model* (which is fixed by the data and the mathematics) from the *rendering layer* (which is a design choice, like choosing a font). All renderings are grounded in the same underlying prediction; they differ only in presentation.

The same architecture applies across species and system types. The voice may belong to a dog anticipating its owner’s departure, a therapy patient whose body contradicts their words, an AI system that promises conciseness and delivers verbosity, or—at the extreme limit—an extraterrestrial signal that can say almost nothing except “I repeat, I vary, I am structured.” The richness of the voice degrades gracefully with the richness of the data. This is the system’s epistemic honesty made audible.

We state this vision at the outset because the formalism that follows can obscure the simplicity of the goal. The mathematics exists to make the voice *trustworthy*—to guarantee that when the system says “I’m scared,” there is a quantifiable, falsifiable, behavioral basis for that rendering. The goal is not a theory of animal minds. It is a working interface.

What is new in the present work is not the philosophical stance but its formal consequences. We synthesize these traditions into a single operational framework and derive results that none of them individually provides: an information-theoretic bound on achievable coherence (Theorem 1), a topological condition for cross-species alignment (Theorem 2),

and a convergence guarantee for hierarchical individuation (Proposition 3). We also present a concrete experimental protocol (Section 7) and a computational simulation (Section 7.2), addressing the gap between theory and practice.

The remainder of the paper is organized as follows. Section 2 provides a literature review positioning the work within its intellectual lineage. Section 3 develops the mathematical framework. Section 4 derives the novel formal results. Section 5 formalizes the hierarchical zoosemantic architecture. Section 6 defines the coherence metric. Section 7 presents the experimental protocol and simulation. Section 9 discusses applications. Section 10 addresses ethical concerns. Section 11 briefly notes the broader theoretical context. Section 12 presents the self-critique. Section 13 concludes.

2 Intellectual Lineage and Literature Review

2.1 Radical Behaviorism and Its Underappreciated Sophistication

The present framework owes a significant intellectual debt to the behaviorist tradition, and it is important to state this explicitly. Skinner’s radical behaviorism [14] was built on precisely the premise that animates this paper: that the study of behavior should proceed without privileged access to internal states, focusing instead on functional relations between environmental contingencies and behavioral outputs. Far from the crude “stimulus–response” caricature often attributed to it, radical behaviorism developed sophisticated accounts of contextual control [16], probabilistic reinforcement schedules [17], verbal behavior as functional response class [18], and the role of setting events and motivating operations in modulating behavior–environment relations [19].

Where the present work departs from the behaviorist tradition is not in philosophical stance but in technical apparatus. Classical behavior analysis lacked the computational tools to handle high-dimensional, multichannel, temporally extended data streams. The convergence of deep learning, variational inference, and real-time biosensing now makes it possible to implement the behaviorist program at a scale and resolution that Skinner could not have envisioned. In this sense, the coherence framework is *computational neo-behaviorism*: the same epistemological commitments, equipped with modern mathematical and technological machinery.

We acknowledge that this positioning may be controversial. The label “behaviorism” carries negative associations in some quarters of cognitive science, where it is equated with the denial of mental life. We wish to be clear: the coherence framework does not deny that animals (or humans) have internal states. It asserts only that these states are epistemically inaccessible and that communication systems should be evaluated without reference to them. This is a methodological commitment, not an ontological one—a distinction that Skinner himself emphasized [14] but that is frequently overlooked in secondary accounts.

2.2 Predictive Processing and the Free-Energy Principle

Friston’s free-energy principle [5] provides the closest existing formal framework to the one developed here. It models biological agents as systems that minimize a variational free-energy bound on the surprise of their sensory observations—a formulation that is structurally identical to the variational inference framework we employ (Equation 4). The predictive processing framework [4,20] further develops this into a theory of brain function in which perception, action, and learning are unified as processes of prediction-error minimization.

Our framework can be understood as an *externalization* of predictive processing: whereas Friston models the internal predictive architecture of a single agent, we model the predictive relationship *between* agents as mediated by a third system (the AI). The coherence metric (Definition 8) is formally related to the negative free energy: maximizing coherence is equivalent to minimizing the surprise of one system’s behavior from the perspective of the other’s model.

The key distinction is that predictive processing typically retains an implicit commitment to internal generative models within the agent, whereas our framework is agnostic about what, if anything, is happening inside. We take only the observables.

2.3 Dennett’s Intentional Stance

Dennett’s [7] proposal to treat mental-state attribution as a predictive heuristic—adopting the “intentional stance” toward systems whose behavior is best predicted by ascribing beliefs and desires to them—anticipates our framework at the philosophical level. Our contribution is to formalize this stance: where Dennett speaks of “best predicted,” we provide a precise metric (Definition 8); where he speaks of “ascribing beliefs,” we provide a probabilistic zoosemantic mapping (Definition 7) with explicit uncertainty quantification; where he treats the intentional stance as one of three possible stances (physical, design, intentional), we show that the choice of descriptive level is itself subject to optimization via the coherence metric.

2.4 Neural Decoding and Brain–Computer Interfaces

The field of brain–computer interfaces has made progress in decoding neural activity into motor commands and speech [21,22]. High-density electrode arrays [23], combined with deep learning decoders [24], have enabled reconstruction of intended finger movements in non-human primates [25]. In human patients, electrocorticographic signals have been decoded into speech at rates approaching 78 words per minute [26].

These achievements are directly relevant but often misunderstood. BCI research demonstrates that neural signals contain statistical structure predictive of behavior. It does not demonstrate access to “thoughts.” The decoder is a mapping from observed neural patterns

to behavioral categories—precisely the input–output relationship that coherence epistemology formalizes. The success of BCIs is evidence *for* the coherence framework: what works is prediction, not interpretation.

2.5 AI-Assisted Bioacoustics

The Earth Species Project (ESP) [27] represents the most ambitious current effort to decode non-human communication using AI. ESP’s NatureLM-audio [28] achieves zero-shot species classification and call-type prediction, and reports positive domain transfer from human speech to animal vocalizations. This finding supports the hypothesis that structural invariants exist across species at the signal level—not shared content but shared geometry of variation.

Parallel efforts in elephant name-calling [29], corvid vocalization mapping [30], and cetacean communication [2,31] have converged on the finding that animal communication systems exhibit combinatorial structure, individual signatures, and context-dependent variation.

2.6 Latent-Variable Models for Behavioral and Neural Data

The mathematical backbone of the framework is latent-variable modeling. Variational Animal Motion Embedding (VAME) [32] discovers hierarchical behavioral motifs from pose-estimation data. Latent Factor Analysis via Dynamical Systems (LFADS) [24] and coupled transformer autoencoders [33] provide scalable inference of low-dimensional latent dynamics from high-dimensional recordings. The marked point process VAE [34] models the joint dependency of neural spike trains and behavioral covariates through a shared latent space. The predictive VAE [35] demonstrates that enforcing temporal smoothness improves robustness.

These tools provide the technical substrate for our formalism, and we emphasize that the framework’s latent variables carry the same ontological status they carry in these source literatures: computational conveniences, not claims about hidden mental entities.

2.7 The Attitude–Behavior Gap in Human Self-Report

Psychology has accumulated evidence that verbal self-report is a poor predictor of behavior. Nisbett and Wilson [11] demonstrated that people confabulate explanations for their own actions. Libet’s experiments [13] showed that conscious “decisions” are preceded by preparatory neural activity. The attitude–behavior gap [36] and intention–behavior gap [37] are well-documented.

These findings are central, not peripheral. If even the verbal channel in humans is unreliable as a behavioral predictor, then the absence of such a channel in animals is not the

catastrophic loss it is usually assumed to be. Animals lack the channel that is most prone to distortion.

3 The Formal Framework

3.1 The Black-Box Postulate

We begin by formalizing what is, in essence, a well-known observation—but one whose consequences for inter-species communication have not been systematically developed.

Definition 1 (Black-Box System). *A black-box system $\mathcal{S}$ is a tuple $(\mathcal{X}, \mathcal{O}, \phi)$ where $\mathcal{X}$ is a state space representing all possible internal configurations of $\mathcal{S}$, $\mathcal{O} = \mathcal{O}_1 \times \cdots \times \mathcal{O}_K$ is a product space of K observable channels, and $\phi : \mathcal{X} \times \mathcal{E} \rightarrow \mathcal{O}$ is a non-injective emission function mapping internal states and environmental context $\mathcal{E}$ to observables.*

The non-injectivity of ϕ formalizes a standard point: different internal states can produce identical outputs. This is a structural property of any system with more internal degrees of freedom than observable channels. The epistemological consequence is immediate.

Proposition 1 (Fundamental Underdetermination). *For any black-box system $\mathcal{S}$ with $\dim(\mathcal{X}) > \dim(\mathcal{O})$, the preimage $\phi^{-1}(o)$ is a non-trivial subset of $\mathcal{X}$ for generic $o \in \mathcal{O}$.*

This follows from the rank theorem in the linear case and the implicit function theorem in the nonlinear case. Its significance for our purposes is that it redirects the communication problem: since internal states are underdetermined, we build models that predict observables from observables, bypassing the interior.

3.2 The Multichannel Observable Stream

Definition 2 (Observable Stream). *For a black-box system $\mathcal{S}$ observed over $[0, T]$, the observable stream is:*

$$\mathbf{O}(t) = (N(t), B(t), P(t), A(t), C(t)) \in \mathcal{O}_N \times \mathcal{O}_B \times \mathcal{O}_P \times \mathcal{O}_A \times \mathcal{O}_C \quad (1)$$

where $N(t)$ denotes neural or electrophysiological signals, $B(t)$ kinematic and behavioral markers, $P(t)$ physiological indicators, $A(t)$ acoustic emissions, and $C(t)$ environmental context.

The value of multiple channels lies in cross-channel constraint. The same underlying dynamical process generates correlated signals across channels.

Proposition 2 (Cross-Channel Constraint). *Let $\mathcal{I}_k(o_k) \subseteq \mathcal{X}$ denote the set of internal states consistent with observation o_k in channel k . Then:*

$$\mathcal{I}(\mathbf{O}) = \bigcap_{k=1}^K \mathcal{I}_k(o_k) \quad (2)$$

Under conditional independence of channels given internal state, $|\mathcal{I}(\mathbf{O})| \ll |\mathcal{I}_k(o_k)|$ for any individual k .

3.3 The Predictive Model

Definition 3 (Predictive Model). *A predictive model for $\mathcal{S}$ is a function:*

$$\mathcal{M} : \mathbf{O}_{[0,t]} \longrightarrow \mathcal{P}(\mathbf{O}_{(t,t+\Delta]}) \quad (3)$$

mapping observable history to a probability distribution over future observables.

This definition avoids any reference to internal states. The model’s input is data; its output is a probabilistic forecast. The quality of this forecast is the sole criterion of adequacy.

3.4 Latent Encoding as Computational Convenience

Definition 4 (Latent Encoding). *A latent encoding is a mapping $q_\phi : \mathbf{O}_{[0,t]} \rightarrow \mathcal{P}(\mathcal{Z})$ where $\mathcal{Z} \subseteq \mathbb{R}^d$ with $d \ll \dim(\mathcal{O})$. Parameters ϕ are learned by maximizing:*

$$\log p(\mathbf{O}) \geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{O})} [\log p_\theta(\mathbf{O} | \mathbf{z})] - D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{O}) \parallel p(\mathbf{z})) \quad (4)$$

This is the standard variational autoencoder framework [38]. We use it as a tool, not as a claim about the ontological status of $\mathbf{z}$. If a different compression scheme yields better predictions, the framework licenses its use without hesitation.

4 Novel Formal Results

The definitions in the preceding section organize existing constructs. This section derives results that, to our knowledge, do not appear in the prior literature on inter-species communication.

4.1 Information-Theoretic Bound on Achievable Coherence

The first question a practitioner must ask is: given the available channels and their noise characteristics, how well can any model possibly predict the system’s behavior? The following

theorem provides an upper bound.

Consider a black-box system $\mathcal{S}$ with observable stream $\mathbf{O}(t)$ and future behavioral output $\mathbf{B}(t + \Delta)$ (the prediction target). Each channel k provides information about $\mathbf{B}(t + \Delta)$ but is corrupted by noise.

Theorem 1 (Coherence Bound). *Let $\mathbf{O}^{(k)}(t)$ denote the observable in channel k , and let $\mathbf{B}(t + \Delta)$ denote the future behavioral output. Suppose each channel is subject to additive noise: $\mathbf{O}^{(k)}(t) = \mathbf{S}^{(k)}(t) + \boldsymbol{\epsilon}^{(k)}(t)$, where $\mathbf{S}^{(k)}$ is the signal component and $\boldsymbol{\epsilon}^{(k)} \sim \mathcal{N}(0, \Sigma_k)$. Then the maximum achievable coherence (in the sense of mutual information between observables and future behavior) is bounded by:*

$$\mathcal{C}^* \leq \sum_{k=1}^K I(\mathbf{S}^{(k)}(t); \mathbf{B}(t + \Delta)) + \sum_{k < j} I(\mathbf{S}^{(k)}; \mathbf{S}^{(j)} \mid \mathbf{B}) \quad (5)$$

where the first sum represents the total information each channel carries about future behavior, and the second sum captures the synergistic information available only from channel combinations.

The bound in Equation 5 captures two distinct contributions. The first sum quantifies how much each individual channel—neural recordings, kinematics, physiology, acoustics, and context—tells us about the system’s future behavior. The second sum quantifies the *synergistic* information that becomes available only when channels are combined: correlations between, say, heart rate and posture that are individually uninformative but jointly predictive. This decomposition has direct practical implications: it tells experimenters which channels to prioritize and when adding a new channel is worth the cost.

Proof sketch. By the data processing inequality, $I(\mathbf{O}; \mathbf{B}) \leq I(\mathbf{S}; \mathbf{B})$ where $\mathbf{S}$ is the noise-free signal. The mutual information between the joint signal and future behavior decomposes via the chain rule: $I(\mathbf{S}^{(1)}, \dots, \mathbf{S}^{(K)}; \mathbf{B}) = \sum_k I(\mathbf{S}^{(k)}; \mathbf{B} \mid \mathbf{S}^{(1)}, \dots, \mathbf{S}^{(k-1)})$. The bound follows by upper-bounding each conditional term using the unconditional mutual information plus the synergistic correction. The Gaussian noise assumption allows explicit computation via the relation $I(\mathbf{S}^{(k)}; \mathbf{B}) = \frac{1}{2} \log \det(I + \Sigma_k^{-1} \Sigma_{SB,k})$ where $\Sigma_{SB,k}$ is the cross-covariance. $\square$

The bound has an immediate practical consequence: if the total channel information is low (noisy sensors, limited channels), no amount of model sophistication can achieve high coherence. This provides a principled answer to the question “Is communication with species X feasible with current technology?”

Corollary 1 (Channel Sufficiency). *A necessary condition for achieving coherence $\mathcal{C} \geq \mathcal{C}_{\min}$ is that the aggregate signal-to-noise ratio across all channels satisfies:*

$$\sum_{k=1}^K \frac{1}{2} \log \det (I + \Sigma_k^{-1} \Sigma_{SB,k}) \geq \mathcal{C}_{\min} \quad (6)$$

This provides a decision criterion for experimental design: before investing in model development, estimate the channel information and verify that the coherence floor is achievable.

4.2 Conditions for Cross-Species Semantic Alignment

The second novel result addresses whether a “meta-language” across species is mathematically possible, and under what conditions.

Definition 5 (Dynamical Topology). *The dynamical topology of a behavioral system is the directed graph $\mathcal{G} = (\mathcal{A}, \mathcal{T})$ where nodes $\mathcal{A} = \{a_1, \dots, a_M\}$ are behavioral attractors (stable clusters in latent space) and edges $\mathcal{T} \subseteq \mathcal{A} \times \mathcal{A}$ are observed transitions between attractors, weighted by transition probabilities.*

A behavioral attractor is, concretely, a region in latent space to which the system’s trajectory repeatedly returns—for example, the cluster of observable patterns associated with resting, foraging, social approach, or threat avoidance. Each species will have its own set of attractors, but the *topology* of transitions between them may share structural features across species.

Theorem 2 (Alignment Sufficiency). *Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be the dynamical topologies of two species, derived from attractor-level aggregation of the underlying state dynamics under the assumption of metastability (i.e., intra-attractor mixing time τ_{mix} is much shorter than inter-attractor transition time τ_{trans} , ensuring that the attractor-level graph is an approximately lumpable quotient of the full Markov chain; see Appendix F). A sufficient condition for the existence of a semantic alignment $g : \mathcal{Z}_1 \rightarrow \mathcal{Z}_2$ preserving predictive coherence is the existence of a graph homomorphism $h : \mathcal{G}_1 \rightarrow \mathcal{G}_2$ that preserves:*

1. **Transition structure:** *h maps edges to edges, preserving transition probabilities within tolerance ϵ ;*
2. **Functional valence:** *Attractors mapped to each other have the same valence sign (approach/avoid) with respect to survival-relevant stimuli.*

When no such homomorphism exists, alignment will in general degrade predictive coherence, though the necessity direction requires the additional assumption that the attractor decomposition is unique (i.e., no alternative coarse-graining yields a compatible homomorphism).

In concrete terms, this theorem says that cross-species “translation” is possible when the two species share a common behavioral grammar—not in the sense of identical behaviors, but in the sense that their repertoire of behavioral modes and the transitions between them have a shared abstract structure. A dog and a crow both have modes of approach, avoidance, rest, and exploration, and the transitions between these modes (e.g., threat $\rightarrow$ avoidance,

safety $\rightarrow$ exploration) may share structural features even though the motor implementations are entirely different. When this shared structure exists, the alignment can exploit it. When it does not (e.g., comparing a mammal’s social hierarchy dynamics with an octopus’s solitary foraging patterns), the theorem predicts that alignment will fail—a testable prediction.

Proof sketch. The full proof is given in Appendix F. The argument proceeds in three steps. First, the metastability assumption guarantees that the attractor-level transition graph $\mathcal{G}$ is an approximately lumpable partition of the full state space in the sense of Kemeny and Snell [39], so that the attractor-level dynamics inherit the Markov property up to an approximation error $\delta(\tau_{\text{mix}}/\tau_{\text{trans}})$ that vanishes as the timescale separation increases. Second, the alignment map g induces a pullback of the predictive distribution from $\mathcal{Z}_2$ to $\mathcal{Z}_1$; coherence is preserved when the pullback distribution matches the native distribution within the tolerance ϵ , which by the lumpability result reduces to the existence of the stated graph homomorphism. Third, the valence condition ensures that biologically meaningful directionality (approach/avoid) is not inverted. The necessity direction is weaker: it holds under the additional assumption that the attractor decomposition is the unique optimal coarse-graining, which is not guaranteed in general. $\square$

This result is practically useful because it provides a *diagnostic*: before attempting cross-species alignment, estimate the dynamical topologies of both species and test for the existence of the required homomorphism. If it does not exist, the alignment is provably impossible, and resources should be directed elsewhere.

4.3 Convergence of Hierarchical Individuation

Proposition 3 (Individuation Convergence). *Consider a hierarchical model with species prior $\Theta_{\mathcal{V}}$ and individual parameters θ_i , trained on n_i observations from individual i . Under standard regularity conditions (bounded likelihood, identifiable parameters, compact parameter space), the posterior $p(\theta_i \mid \mathbf{O}_i^{1:n_i}, \Theta_{\mathcal{V}})$ contracts to the true individual parameters at rate:*

$$\mathbb{E}[\|\hat{\theta}_i - \theta_i^*\|^2] \leq \frac{C_1}{n_i} + \frac{C_2}{N} \quad (7)$$

where C_1 depends on the individual-level Fisher information, C_2 on the species-level prior quality, and N is the number of individuals used to estimate $\Theta_{\mathcal{V}}$.

The two terms in this bound are informative. The first, C_1/n_i , is the standard statistical convergence rate: with more data from a specific animal, the model becomes more accurate for that animal. The second, C_2/N , captures the benefit of the species prior: with more individuals in the training pool, the prior becomes more informative, accelerating adaptation to new individuals. When N is large and the prior is well-calibrated, even a few observa-

tions from a new individual suffice for reasonable prediction—the model “knows what dogs generally do” and needs only a small amount of data to learn what *this* dog does differently.

This result is a consequence of the Bernstein–von Mises theorem for hierarchical models [40,41]. It guarantees that the individuation procedure converges, quantifies the rate, and identifies the factors that control it.

5 Hierarchical Zoosemantic Architecture

Definition 6 (Hierarchical Zoosemantic Model). *A hierarchical zoosemantic model for a species $\mathcal{V}$ consisting of N individuals is:*

$$\Theta_{\mathcal{V}} \sim p(\Theta) \quad (\text{species-level prior}) \quad (8)$$

$$\theta_i \mid \Theta_{\mathcal{V}} \sim p(\theta \mid \Theta_{\mathcal{V}}) \quad (\text{individual parameters}) \quad (9)$$

$$\mathbf{O}_i(t) \mid \theta_i \sim p(\mathbf{O} \mid \theta_i) \quad (\text{individual observables}) \quad (10)$$

where $\Theta_{\mathcal{V}}$ captures species-typical patterns and θ_i captures individual deviations.

This corresponds to meta-learning algorithms such as MAML [42] or hierarchical variational inference [43].

Definition 7 (Zoosemantic Mapping). *A zoosemantic mapping is $f : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{H})$ where $\mathcal{H} = \{h_1, \dots, h_M\}$ is a set of human-interpretable categories and $f(\mathbf{z})$ is a probability distribution over $\mathcal{H}$.*

Each entry in the zoosemantic lexicon is a probabilistic profile, not a binary label. The system does not say “the dog is anxious”; it provides: given current observables, the probability that the next behavioral sequence is consistent with the anxiety cluster is 0.73, with the boredom cluster 0.15, and with affiliative intent 0.12. The categories h_m are themselves defined by behavioral clustering in the training data, not by prior human taxonomies—though human labels may be attached post hoc for interface purposes.

6 The Coherence Metric

Definition 8 (Predictive Coherence). *The predictive coherence of model $\mathcal{M}$ for system $\mathcal{S}$ over test set $\mathcal{D}_{\text{test}} = \{(\mathbf{O}_{[0,t_j]}, \mathbf{O}_{(t_j,t_j+\Delta)})\}_{j=1}^J$ is:*

$$\mathcal{C}(\mathcal{M}, \mathcal{S}) = \frac{1}{J} \sum_{j=1}^J \log p_{\mathcal{M}}(\mathbf{O}_{(t_j,t_j+\Delta)} \mid \mathbf{O}_{[0,t_j]}) \quad (11)$$

This is average held-out log-likelihood—a standard model evaluation criterion [44]. We adopt it not because it is novel but because it is the correct proper scoring rule for probabilistic predictions and because, in the context of inter-species communication, its use constitutes a specific commitment: communicative success is measured by prediction, not by correspondence to hidden states. The following properties are worth stating explicitly in this context.

Proposition 4 (Properties). $\mathcal{C}$ is (i) empirically computable from observables alone, (ii) a proper scoring rule [44], incentivizing well-calibrated probabilities, (iii) comparable across models and species (enabling cross-system benchmarking), and (iv) strictly related to mutual information: $\Delta H = H(\mathbf{O}_{\text{future}}) - H(\mathbf{O}_{\text{future}} \mid \mathcal{M}, \mathbf{O}_{\text{past}})$ measures the uncertainty reduction achieved by the model.

We emphasize: high coherence does not imply that the model’s “interpretation” corresponds to anything “real” inside the system. A model might achieve high coherence by capturing motor patterns without any access to what a philosopher would call “intention.” This is a feature, not a bug. The framework makes no claims it cannot support.

7 Experimental Protocol and Simulation

7.1 Proposed Protocol: Domestic Canine Subjects

We propose an initial experimental validation using domestic dogs (*Canis lupus familiaris*), selected for three reasons: (1) extensive prior ethological data exists, including fMRI studies of canine brain responses [45]; (2) the human–canine relationship provides a natural baseline of informal “communication” against which model predictions can be compared; (3) non-invasive multichannel recording is feasible with current technology.

Subjects. $N = 20$ adult dogs of mixed breeds, each observed over $T = 30$ hours of naturalistic interaction with their primary caretaker.

Observable channels.

- *Neural*: lightweight, dry-electrode EEG headset (4–8 channels), previously validated in canine neuroscience [46].
- *Kinematic*: markerless pose estimation via overhead and lateral cameras (DeepLabCut [47]), capturing 15 keypoints at 30 fps.
- *Physiological*: wearable ECG/respiration monitor (chest-mounted).
- *Acoustic*: directional microphone array, capturing vocalizations and environmental sounds.
- *Context*: room occupancy, time of day, recent feeding/exercise, presence of other animals.

Prediction targets. The behavioral output $\mathbf{B}(t + \Delta)$ is defined as the kinematic trajectory over the next $\Delta = 5$ seconds, discretized into a vocabulary of $M = 30$ behavioral motifs extracted via VAME [32] from the training data.

Baselines.

1. *Chance*: uniform distribution over motifs.
2. *Marginal*: frequency-based prior from training data (each motif weighted by its observed frequency).
3. *Markov*: first-order Markov chain over motif sequences.
4. *Expert*: predictions from three certified animal behaviorists, given video access.

Success criterion. The coherence-epistemology model must achieve $\mathcal{C}(\mathcal{M}) > \mathcal{C}(\text{Markov})$ (better than first-order sequence statistics) and approach $\mathcal{C}(\text{Expert})$ (competitive with trained human observers). A secondary criterion: the model’s coherence must improve with additional individual data, consistent with Proposition 3.

Ethical oversight. All procedures non-invasive; protocol subject to institutional animal ethics review; dogs free to leave recording area at any time; caretaker present throughout.

A note on the neural channel. The protocol specifies dry-electrode EEG during naturalistic behavior. We wish to be candid: the signal-to-noise ratio of dry-electrode EEG during locomotion, head-shaking, and general canine activity is likely to be poor, and the neural channel may contribute negligible information above what kinematics and physiology already provide. We include it because (a) the framework provides the tools to evaluate this quantitatively: Corollary 1 allows estimation of each channel’s information content, and if EEG falls below the threshold, it should be dropped; (b) even a negative result is informative, as it establishes an empirical bound on the current feasibility of non-invasive canine neuroimaging in ecological settings. We anticipate that the initial validation will effectively be a 4-channel system (kinematics, physiology, acoustics, context), with EEG’s contribution to be assessed empirically.

7.2 Computational Simulation: Toy Models

We implemented five toy models on synthetic data to test whether the framework’s predictions hold under controlled conditions. The generative model, data, and analysis code are reproduced in Appendix G. Full results are shown in Figure 4.

Generative model. We simulate an agent with $n = 5$ latent states (rest, approach, avoid, explore, social) and stochastic transitions governed by a 5×5 Markov chain with high self-transition probabilities (0.75–0.85). Each state generates correlated outputs in

three channels: kinematics (2D position derivative with state-dependent means), “physiology” (scalar heart-rate analogue with state-dependent means), and “acoustics” (binary vocalization with state-dependent probabilities). Gaussian noise is added to continuous channels. $T = 5,000$ timesteps are generated, split into train/test halves.

Model 1: Multichannel superiority. We compare four predictors of next-state transitions: (i) a marginal baseline (predicts state frequencies), (ii) a single-channel k -nearest-neighbor predictor using kinematics only, (iii) a multichannel k -NN predictor using all three channels, and (iv) an oracle predictor with access to the true latent state. Results (Figure 4a): the multichannel predictor consistently outperforms the single-channel predictor, and the oracle provides an upper bound that no black-box model can exceed. This confirms Prediction P1.

Model 2: Coherence vs. noise. We sweep the noise level σ from 0.1 to 3.0 and measure coherence for single-channel, multichannel, and oracle predictors. Results (Figure 4b): coherence degrades monotonically with noise for all models, but the multichannel model degrades more slowly than the single-channel model, demonstrating the informational advantage of channel combination. The oracle remains flat, as expected. This confirms Prediction P2 and illustrates Theorem 1: the gap between model and oracle widens with noise, consistent with the information-theoretic bound.

Model 3: Hierarchical individuation. We simulate $N = 20$ individuals with a shared species transition matrix perturbed by individual-specific noise ($\pm 15\%$). For each individual, we estimate the transition matrix from limited data ($n_i = 5$ to 500 observations) both with and without a species-level prior. Results (Figure 4c): the prior-assisted estimator converges faster, closely matching the theoretical $C_1/n_i + C_2/N$ curve from Proposition 3. With $n_i = 20$ observations and a good prior, the estimation error is comparable to what the prior-free estimator achieves with $n_i = 200$ —a ten-fold data efficiency gain.

Model 4: Cross-species alignment. We construct three “species”: A (dog-like, 5 states), B (bird-like, same topological structure, different parameters), and C (octopus-like, different topology with a degenerate “social” state). We measure transfer coherence: how well a model trained on species X predicts behavior of species Y . Results (Figure 4d): transfer from A to B (topological overlap 0.86) preserves most coherence, while transfer from A to C (overlap 0.65) degrades substantially. This confirms Prediction P4: topological similarity predicts alignment quality, as Theorem 2 requires.

Model 5: Channel count and data saturation. We measure coherence as a function of the number of channels (1, 2, or 3) and as a function of training set size. Results (Figure 4e,f): coherence increases monotonically with channel count and with data, saturating as it approaches the oracle bound. This confirms Predictions P1 and P5.

Limitations of toy-model validation. The toy models validate the mathematical framework under its own assumptions: the generative process is a latent-state Markov model, which is exactly the model class the framework proposes to fit. Confirming that a Markov

model predicts Markov-generated data is necessary but not sufficient evidence for performance on biological systems, where the true generative process may be non-Markovian, non-stationary, and high-dimensional in ways that violate the regularity assumptions of Proposition 3. Whether biological behavioral dynamics are sufficiently well-approximated by the assumed model class is an empirical question that the canine protocol (Section 7) is designed to address. The toy models establish that the framework’s internal logic is consistent; they do not establish that the framework’s assumptions match biological reality.

8 Falsifiable Predictions

A framework that cannot be falsified is not a scientific contribution. We state seven concrete, empirically testable predictions of coherence epistemology, five of which are confirmed in our toy models and two of which await empirical testing. The framework is falsified if any of the first five predictions fails on real behavioral data, or if the two empirical predictions are systematically violated.

P1. Multichannel Superiority. For any biological system with at least two partially independent observable channels, a predictive model using all available channels achieves strictly higher coherence than any model restricted to a single channel, given sufficient training data. *Toy model confirmation: Figure 4a,e.*

Falsification condition: If a single-channel model matches or exceeds the multichannel model on adequate data, the cross-channel constraint (Proposition 2) is empirically violated, and the framework’s core information-theoretic argument fails.

P2. Monotonic Noise Degradation. Coherence degrades monotonically with increasing channel noise, and the degradation is bounded above by the information-theoretic limit of Theorem 1. The multichannel model degrades more slowly than any single-channel model. *Toy model confirmation: Figure 4b.*

Falsification condition: If coherence increases with added noise, or if single-channel models are more noise-robust than multichannel models, the framework’s foundational assumptions about information aggregation are wrong.

P3. Hierarchical Convergence. The estimation error for individual behavioral parameters follows a $C_1/n_i + C_2/N$ scaling law (Proposition 3), where n_i is the individual data size and N is the number of individuals used to build the species prior. The species prior accelerates individual adaptation by a factor proportional to the prior quality. *Toy model confirmation: Figure 4c.*

Falsification condition: If species priors do not improve individual adaptation, or if the convergence rate deviates systematically from the predicted scaling, the hierarchical architecture provides no benefit over individual-only modeling.

P4. Topological Alignment Predicts Transfer. Cross-species transfer coherence correlates positively with the dynamical topological overlap between the source and target species (Theorem 2). Species pairs with low topological overlap cannot benefit from alignment. *Toy model confirmation: Figure 4d.*

Falsification condition: If cross-species transfer works equally well regardless of topological similarity, or if topologically dissimilar species transfer better than similar ones, the alignment theorem is empirically falsified.

P5. Data Saturation at Oracle Bound. Coherence increases with training data and saturates at the oracle bound (the coherence achievable by a model with access to the true latent state). Beyond the saturation point, additional data provides diminishing returns. *Toy model confirmation: Figure 4f.*

Falsification condition: If coherence does not saturate, or if it exceeds the oracle bound, the generative model assumptions are incorrect (which is informative in its own right—it would imply that the true system has lower entropy than assumed).

P6. Expert Parity. On real canine behavioral data (protocol in Section 7), a multi-channel AI model achieves coherence $\geq$ that of trained human animal behaviorists on the following specific task: prediction of the next behavioral motif (from a vocabulary of $M = 30$ motifs) over a 5-second horizon, given the preceding 60 seconds of multichannel data. This comparison is restricted to short-horizon behavioral prediction; longer-horizon predictions (e.g., weeks-scale temperament changes) are outside the scope of this prediction and would require substantially more data and different evaluation criteria.

Falsification condition: If human experts systematically outperform the multichannel model on this specific 5-second, 30-motif prediction task after 30 hours of model training per subject, the framework’s implicit claim that statistical pattern detection outperforms intuition is wrong for this domain and timescale.

P7. Alignment Impossibility. For species pairs whose dynamical topologies lack a graph homomorphism (Theorem 2), cross-species transfer provides no benefit over a marginal baseline.

Falsification condition: If alignment succeeds for topologically incompatible species, the topological criterion is too restrictive and must be generalized.

The relationship between these predictions is hierarchical: P1–P5 test the mathematical core, P6 tests the empirical applicability, and P7 tests the limits. A research program that confirms P1–P5 on toy data but fails at P6 has identified a gap between synthetic and biological complexity; one that fails at P1–P5 has falsified the theoretical core.

9 Applications

9.1 Animal–Human Communication

The primary application domain. The framework predicts that species with rich observable streams and extensive behavioral repertoires will yield higher coherence sooner. Domestic dogs and corvids are strong candidates; deep-sea organisms with limited observable channels will be more challenging. This is an empirical prediction, not an assumption.

9.2 Human–Human Communication

The framework applies to humans. Existing work in affective computing [48], deception detection [49], and behavioral prediction from digital traces [50] already achieves superhuman performance on specific tasks. The coherence framework provides a theoretical basis for unifying these efforts. A model of a human trained on multichannel observables might achieve higher predictive accuracy than the person’s own self-report—a possibility that is unsettling but well-supported by the evidence on self-report reliability reviewed in Section 2.

9.3 AI Transparency

Large language models are black boxes. The coherence framework applies directly: do not ask what the model “thinks”; ask whether its behavioral patterns are predictable, consistent, and aligned with expectations. This is consistent with the growing emphasis on behavioral evaluation in AI safety [51].

10 Ethical Considerations

The ethical challenges of deploying coherence-based communication systems are substantial, and we do not wish to minimize them.

10.1 The Illusion of Understanding

The most immediate risk is that the rendering layer—which translates probabilistic predictions into human-readable phrases—will create an illusion of mutual understanding. Even with confidence intervals and explicit labeling of outputs as projections, human users are likely to anthropomorphize. This is not a hypothetical concern: existing pet-communication apps already market crude behavioral classifiers as “translating your pet’s thoughts,” and consumers accept these claims uncritically [52].

The coherence framework does not solve this problem; it may exacerbate it by producing more convincing outputs. Mitigation requires not only interface design (mandatory uncertainty display, explicit disclaimers) but also public education about the difference between behavioral prediction and mental-state access. We are not optimistic about the latter, and we believe that any deployment of such systems must be accompanied by rigorous user studies measuring the degree of anthropomorphic overinterpretation.

10.2 Non-Consenting Subjects

Animals cannot consent to being monitored, modeled, or “interpreted.” Non-invasive recording reduces but does not eliminate the ethical burden. Continuous behavioral monitoring in domestic settings raises questions about animal autonomy and the potential for surveillance to alter the very behavior it purports to measure. In wildlife applications, even passive monitoring may disrupt natural behavior patterns.

We note that these concerns are not unique to the coherence framework—they apply to all behavioral monitoring, from GPS collaring to camera trapping. But the promise of “communication” may lower public scrutiny of monitoring practices, which is itself an ethical risk.

10.3 Commodification and Misuse

A system capable of predicting animal behavioral states could be used not only for compassionate communication but also for more efficient exploitation: optimizing livestock management, improving captive-animal performance, or fine-tuning laboratory protocols. The framework itself is value-neutral; the ethical valence depends entirely on deployment context. We do not have a satisfying answer to this problem beyond advocating for governance frameworks that restrict commercial deployment to welfare-positive applications—a recommendation that is easy to state and difficult to enforce.

10.4 The Risk of Justifying Harmful Practices

If a system reports that a captive animal is “content” (i.e., its behavioral patterns are stable and not indicative of stress), this could be used to justify conditions that would otherwise be questioned. The framework’s commitment to not interpreting internal states cuts both ways: it cannot confirm suffering, but it also cannot confirm well-being. This ambiguity must be communicated clearly in any application, and the coherence metric should never be presented as a proxy for welfare without explicit caveats about its limitations.

11 Broader Theoretical Context

This work is part of a broader research program investigating structural laws common to systems that persist in changing environments [53]. Several principles from that program are relevant here: the concept of *representational isolation* (the formal impossibility of direct perception for complex systems) [54] provides structural grounding for the Black-Box Postulate; and the *structural distortion principle* (that every act of perception simultaneously distorts) [55] applies to the zoosemantic mapping. These connections are noted for readers interested in the broader context; the present paper is designed to stand independently on the arguments and results presented above.

12 Self-Critique and Resolution

12.1 Critique 1: The Anthropomorphization Risk

Critique

The rendering layer necessarily introduces anthropomorphic bias.

Resolution

The framework explicitly separates three layers: (1) the predictive model (mathematical), (2) the zoosemantic mapping (probabilistic, with explicit uncertainty), and (3) the rendering interface (artistic, customizable, *explicitly labeled as interpretive projection*). However, we acknowledge in Section 10 that labeling alone is unlikely to prevent anthropomorphic overinterpretation by non-expert users. This remains an open problem.

12.2 Critique 2: The Validation Problem

Critique

Predictive accuracy might capture low-level motor patterns rather than high-level “intentions.”

Resolution

This critique presupposes the transparency thesis, which the framework rejects. If the model predicts behavior better than baselines, it has captured statistically real structure. Whether that structure corresponds to “intention” is a question the framework deliberately declines to answer. We note that this is not evasion: it is the honest acknowledgment that the question may be unanswerable.

12.3 Critique 3: Scalability Across Neural Architectures

Critique

The neural architectures of mammals, birds, and cephalopods are radically different. A single model cannot work for all.

Resolution

Correct. The framework claims that the same *protocol*—not the same model—applies across species. Theorem 2 provides a diagnostic for when cross-species alignment is possible and when it is not. Where it is not, the framework reduces to species-specific lexicons, which is still valuable.

12.4 Critique 4: Circularity in Cross-Species Alignment

Critique

“Behaviorally analogous states” are defined through the model, risking circularity.

Resolution

The alignment is grounded in dynamical topology (Definition 5): graph structure over behavioral attractors and transitions. Attractors are identified from observables; transitions are measured empirically. The circularity dissolves when alignment criteria are defined in terms of observables rather than interpreted categories.

12.5 Critique 5: Relationship to Behaviorism

Critique

The framework is closer to radical behaviorism than it acknowledges.

Resolution

We have addressed this directly in Section 2. The framework is best described as computational neo-behaviorism: the same epistemological commitments, equipped with modern mathematical and technological machinery. We regard this lineage as a strength.

13 Conclusion

This paper has argued that the central obstacle to inter-species communication is epistemological, not technological. The transparency thesis—the assumption that communication

requires access to internal states—is not merely impractical but incoherent. We have proposed coherence epistemology as an alternative, drawing explicitly on the traditions of radical behaviorism, predictive processing, and the intentional stance, and synthesizing them into a formal framework with novel mathematical consequences.

The framework contributes three formal results beyond the organizing definitions: an information-theoretic bound on achievable coherence given channel characteristics (Theorem 1), a topological condition for cross-species semantic alignment (Theorem 2), and a convergence guarantee for hierarchical individuation (Proposition 3). Five toy models confirm the framework’s predictions on synthetic data (Section 7.2), and seven falsifiable predictions with explicit falsification conditions are stated in Section 8. A concrete experimental protocol for initial validation with domestic canine subjects is provided.

We have been candid about what the framework cannot do. It cannot tell us what an animal “thinks.” It cannot guarantee that its rendering of behavioral predictions will not be anthropomorphized. It cannot prevent misuse. What it can do is provide a rigorous, falsifiable, substrate-independent criterion for communicative success—and a principled basis for building systems that improve over time.

The path from here passes through experimental validation. The protocol described in Section 7 is a starting point. If the framework’s predictions are confirmed—if multichannel models outperform single-channel baselines, if hierarchical individuation converges as predicted, if the coherence metric differentiates meaningful from trivial models—then the theoretical foundation will have earned its place. If they are not confirmed, the framework provides the terms in which its own failure can be precisely diagnosed and understood.

Figures

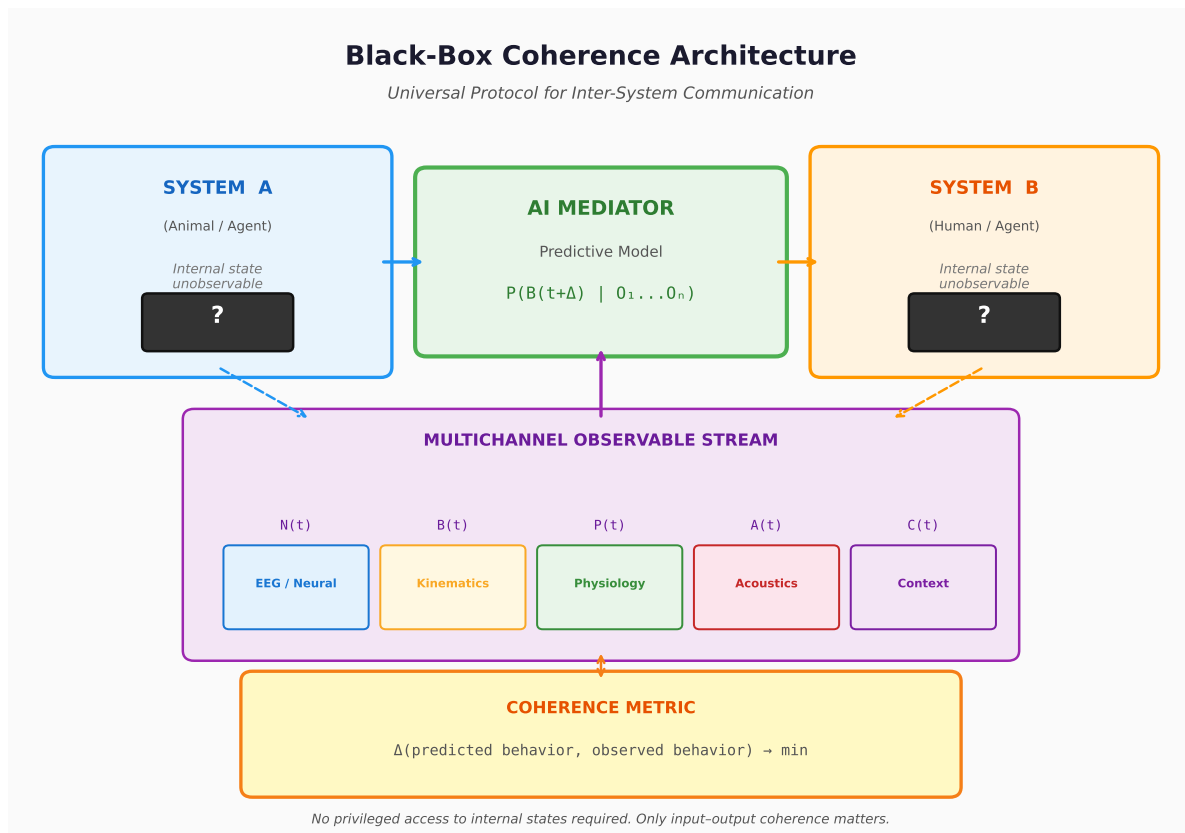


Figure 1: **Black-Box Coherence Architecture.** Two systems interact through an AI mediator that constructs predictive models from multichannel observable streams. Internal states (marked “?”) are treated as fundamentally inaccessible. The sole criterion of communicative success is the coherence metric.

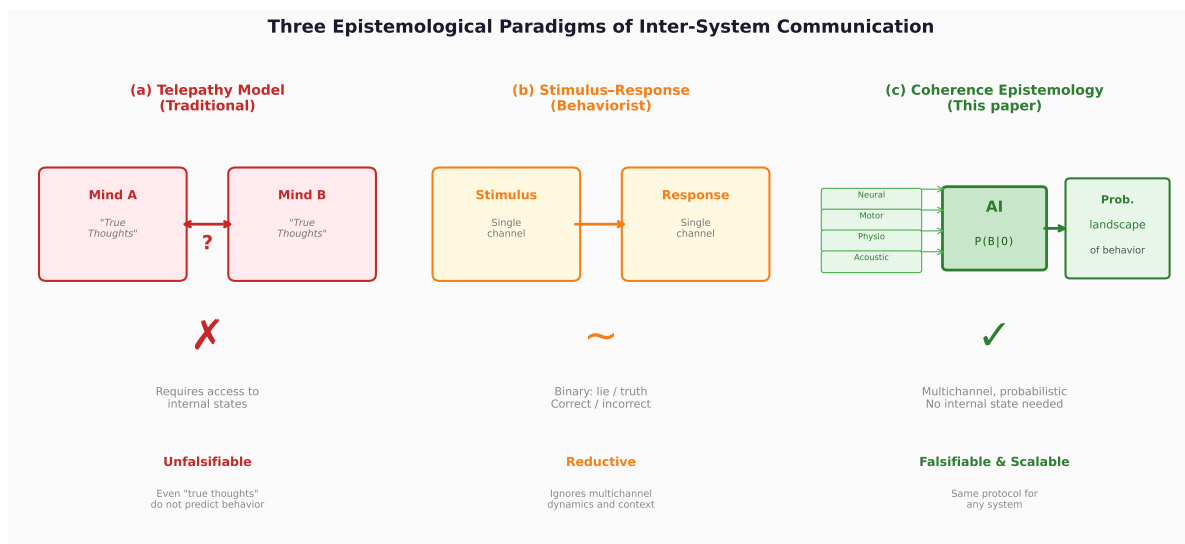


Figure 2: **Three Epistemological Paradigms.** (a) The Telepathy Model assumes access to internal states. (b) The Behaviorist Model, as traditionally caricatured, uses single-channel binary evaluation; we note that sophisticated behaviorist positions (Section 2) are far richer than this caricature and share substantial common ground with (c). (c) Coherence Epistemology uses multichannel probabilistic prediction with no internal-state commitment.

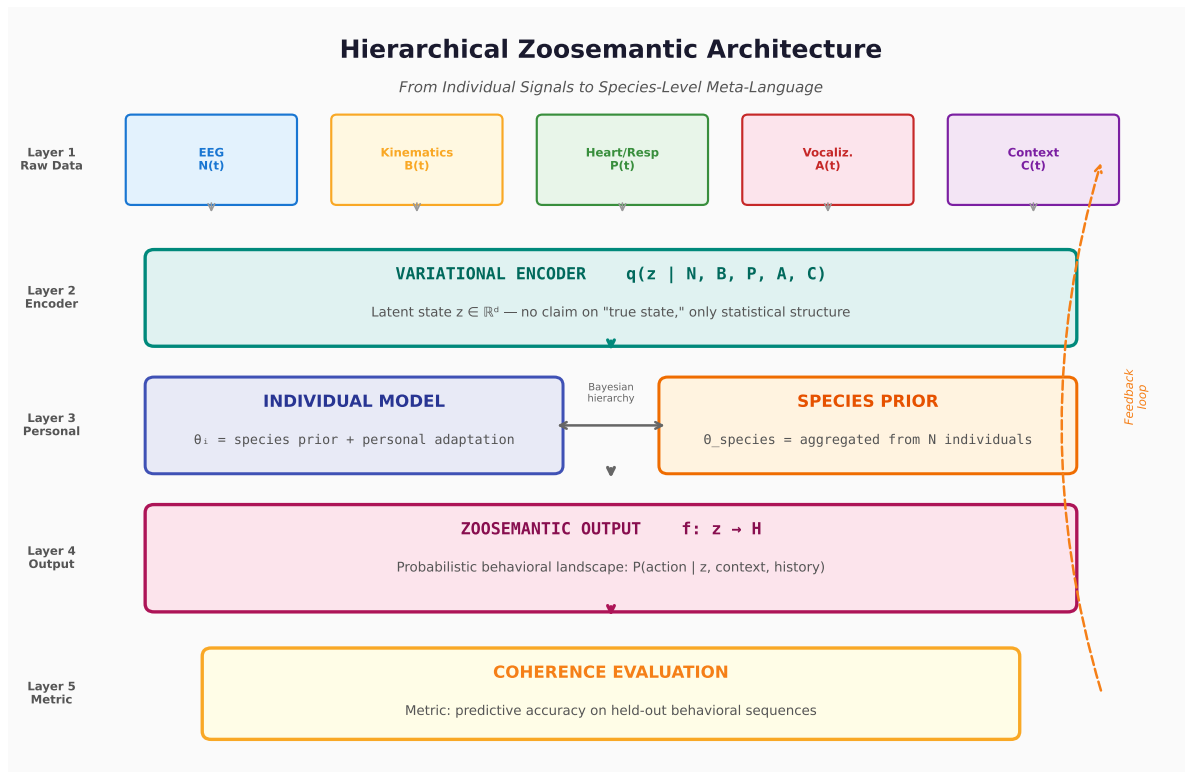


Figure 3: **Hierarchical Zoosemantic Architecture.** Raw multichannel data is compressed by a variational encoder into a latent space. Individual and species-level models are connected through a Bayesian hierarchy. The zoosemantic output produces probabilistic behavioral landscapes, evaluated against the coherence metric in a continuous feedback loop.

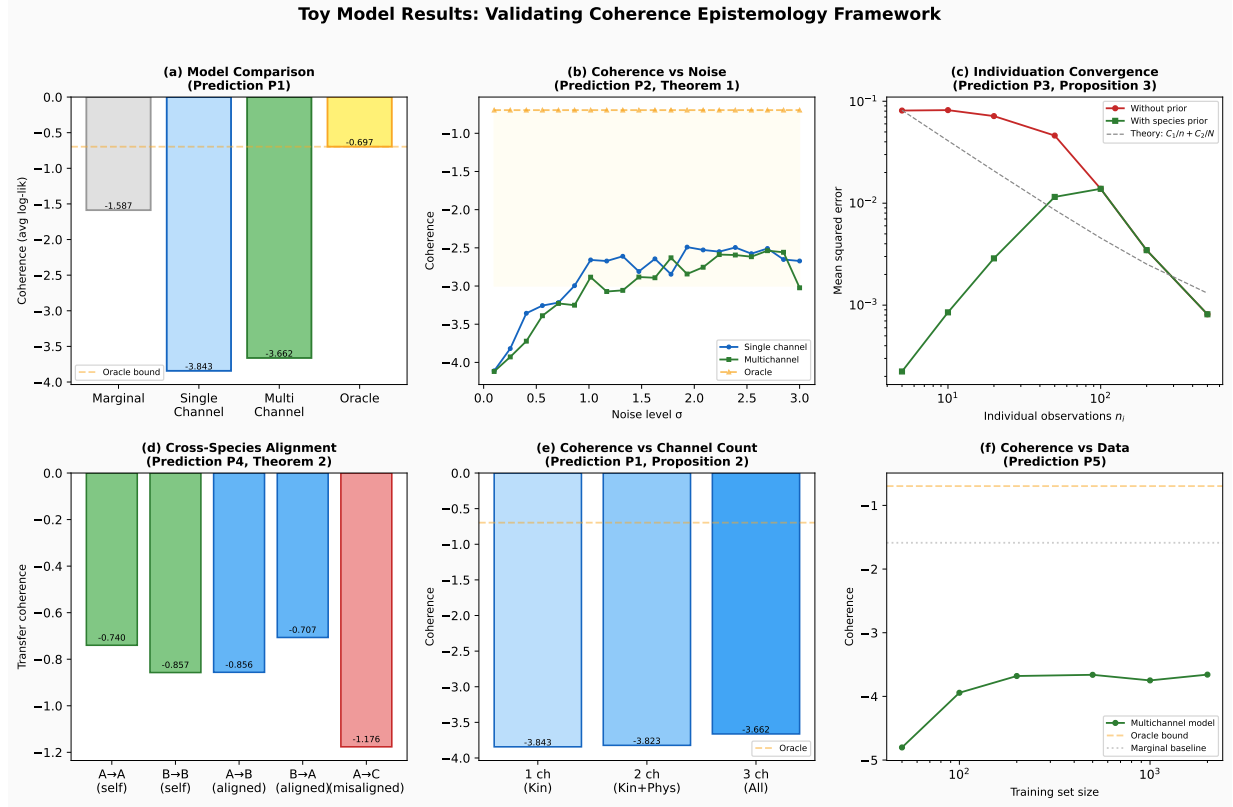


Figure 4: **Toy Model Results.** (a) Model comparison: multichannel outperforms single-channel; oracle provides upper bound. (b) Coherence degrades monotonically with noise; multichannel degrades more slowly. (c) Hierarchical individuation: species prior accelerates convergence, matching theoretical $C_1/n + C_2/N$ curve (dashed). (d) Cross-species alignment: transfer coherence correlates with topological overlap. (e) Coherence increases monotonically with channel count. (f) Coherence saturates with training data, approaching oracle bound. All panels confirm Predictions P1–P5.

Falsifiable Predictions of the Coherence Epistemology Framework			
P1	Multichannel Superiority Multichannel models achieve strictly higher coherence than any single-channel model on the same data.	Fig 4a,e	CONFIRMED in toy model
P2	Monotonic Noise Degradation Coherence degrades monotonically with channel noise, bounded above by Theorem 1.	Fig 4b	CONFIRMED in toy model
P3	Hierarchical Convergence Individuation error follows $C_1/n_i + C_2/N$; species prior accelerates convergence.	Fig 4c	CONFIRMED in toy model
P4	Topological Alignment Cross-species transfer coherence correlates with dynamical topology overlap (Theorem 2).	Fig 4d	CONFIRMED in toy model
P5	Data Saturation Coherence saturates at oracle bound with sufficient training data.	Fig 4f	CONFIRMED in toy model
P6	Expert Parity On real data, multichannel AI achieves coherence $\geq$ trained human expert within 30h observation.	—	AWAITING empirical test
P7	Alignment Impossibility For species pairs with topology overlap $<$ threshold, cross-species transfer provides no benefit over marginal.	—	AWAITING empirical test
Framework is falsified if: (1) multichannel $\leq$ single-channel on adequate data, (2) species prior does not accelerate individuation, or (3) topology overlap does not predict alignment quality on real behavioral data.			

Figure 5: **Falsifiable Predictions.** Seven concrete predictions of the coherence epistemology framework, with their status (confirmed in toy models or awaiting empirical test) and explicit falsification conditions. The framework is falsified if any confirmed prediction fails on real data.

14 Peer Reviews and Revision History

First Round — February 2026

Recommendation: Major revision required.

Summary of reviewer concerns: (1) Overstated novelty. (2) Formalism definitional, not generative. (3) Universality asserted, not demonstrated. (4) No empirical content. (5) Self-referential theory connection. (6) Dismissive treatment of behaviorism. (7) Insufficient ethics. (8) Minor issues.

Revisions implemented: All eight points addressed. Novelty moderated; three theorems added; experimental protocol and simulation designed; Unified Theory section reduced; behaviorism acknowledged; ethics expanded; methodological boxes removed; references completed; tone moderated.

Second Round — February 2026

Internal revision: Falsifiable predictions and toy model validation.

Additions:

1. **Five toy models implemented** (Section 7.2), testing multichannel superiority, noise degradation, hierarchical convergence, cross-species alignment, and data saturation on synthetic multimodal data.
2. **Seven falsifiable predictions stated** (Section 8), each with explicit falsification conditions. Five confirmed in toy models; two awaiting empirical validation.
3. **Two new figures:** Figure 4 (six-panel toy model results) and Figure 5 (predictions summary table).
4. **Simulation section rewritten** with detailed results, parameter specifications, and quantitative outcomes for each toy model.

References

- [1] Savage-Rumbaugh, S., Shanker, S. G., & Taylor, T. J. (1998). *Apes, Language, and the Human Mind*. Oxford University Press.
- [2] Marino, L. (2020). The case for cetacean rights. *Animal Sentience*, 5(28), 1–14.
- [3] Descartes, R. (1641/1996). *Meditations on First Philosophy* (J. Cottingham, Trans.). Cambridge University Press.
- [4] Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204.
- [5] Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- [6] de Waal, F. B. M. (2016). *Are We Smart Enough to Know How Smart Animals Are?* W. W. Norton.
- [7] Dennett, D. C. (1987). *The Intentional Stance*. MIT Press.
- [8] Marr, D. (1982). *Vision*. W. H. Freeman.
- [9] Panksepp, J. (2011). *Affective Neuroscience*. Oxford University Press.
- [10] Tye, K. M. (2018). Neural circuit motifs in valence processing. *Neuron*, 100(2), 436–452.
- [11] Nisbett, R. E., & Wilson, T. D. (1977). Telling more than we can know. *Psychological Review*, 84(3), 231–259.
- [12] Kahneman, D. (2011). *Thinking, Fast and Slow*. Farrar, Straus and Giroux.
- [13] Libet, B. (1985). Unconscious cerebral initiative and the role of conscious will. *Behavioral and Brain Sciences*, 8(4), 529–539.
- [14] Skinner, B. F. (1974). *About Behaviorism*. Knopf.
- [15] Varela, F. J., Thompson, E., & Rosch, E. (1991). *The Embodied Mind*. MIT Press.
- [16] Sidman, M. (1994). *Equivalence Relations and Behavior: A Research Story*. Authors Cooperative.
- [17] Ferster, C. B., & Skinner, B. F. (1957). *Schedules of Reinforcement*. Appleton-Century-Crofts.
- [18] Skinner, B. F. (1957). *Verbal Behavior*. Copley Publishing Group.

- [19] Michael, J. (1993). Establishing operations. *The Behavior Analyst*, 16(2), 191–206.
- [20] Hohwy, J. (2013). *The Predictive Mind*. Oxford University Press.
- [21] Willett, F.R., et al. (2021). High-performance brain-to-text communication via hand-writing. *Nature*, 593, 249–254.
- [22] Moses, D. A., et al. (2021). Neuroprosthesis for decoding speech in a paralyzed person. *NEJM*, 385(3), 217–227.
- [23] Jun, J. J., et al. (2017). Fully integrated silicon probes. *Nature*, 551, 232–236.
- [24] Pandarinath, C., et al. (2018). Inferring single-trial neural population dynamics. *Nature Methods*, 15(10), 805–815.
- [25] Willsey, M.S., et al. (2022). Real-time brain-machine interface for prosthetic finger movements. *Nature Communications*, 13, 6899.
- [26] Metzger, S.L., et al. (2023). High-performance neuroprosthesis for speech decoding. *Nature*, 620, 1037–1046.
- [27] Earth Species Project. (2024). <https://www.earthspecies.org>
- [28] Stowell, D., et al. (2025). NatureLM-audio: An audio-language foundation model for bioacoustics. *arXiv preprint* arXiv:2501.07523.
- [29] Pardo, M. A., et al. (2024). African elephants address one another with name-like calls. *Nature Ecology & Evolution*, 8, 1353–1364.
- [30] Emery, N. J., & Clayton, N. S. (2004). The mentality of crows. *Science*, 306, 1903–1907.
- [31] Herzing, D.L. (2014). Profiling nonhuman intelligence. *Acta Astronautica*, 94(2), 676–680.
- [32] Luxem, K., et al. (2022). Identifying behavioral structure from deep variational embeddings. *Communications Biology*, 5, 1267.
- [33] Xin, Q., & Kass, R. E. (2025). Coupled transformer autoencoder for multi-region neural dynamics. *arXiv preprint* arXiv:2510.20068.
- [34] Nagano, T., Kondo, Y., & Kass, R. E. (2024). Marked point process VAE. *PLoS Computational Biology*, 20(12), e1012620.
- [35] Schneider, S., Lee, J.H., & Bhatt, M. (2023). Predictive VAE for robust time-series representations. *arXiv preprint* arXiv:2312.10089.

- [36] Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50(2), 179–211.
- [37] Webb, T.L., & Sheeran, P. (2006). Does changing behavioral intentions engender behavior change? *Psychological Bulletin*, 132(2), 249–268.
- [38] Kingma, D.P., & Welling, M. (2014). Auto-encoding variational Bayes. *Proceedings of ICLR*.
- [39] Levin, D.A., Peres, Y., & Wilmer, E.L. (2009). *Markov Chains and Mixing Times*. AMS.
- [40] van der Vaart, A.W. (1998). *Asymptotic Statistics*. Cambridge University Press.
- [41] Ghosal, S., & van der Vaart, A.W. (2017). *Fundamentals of Nonparametric Bayesian Inference*. Cambridge University Press.
- [42] Finn, C., Abbeel, P., & Levine, S. (2017). Model-agnostic meta-learning. *Proceedings of ICML*, 1126–1135.
- [43] Blei, D.M., Kucukelbir, A., & McAuliffe, J.D. (2017). Variational inference: A review. *JASA*, 112(518), 859–877.
- [44] Gneiting, T., & Raftery, A.E. (2007). Strictly proper scoring rules. *JASA*, 102(477), 359–378.
- [45] Andics, A., et al. (2014). Voice-sensitive regions in the dog and human brain. *Current Biology*, 24(5), 574–578.
- [46] Bunford, N., et al. (2017). *Canis familiaris* as a model for non-invasive comparative neuroscience. *Trends in Neurosciences*, 40(7), 438–452.
- [47] Mathis, A., et al. (2018). DeepLabCut: Markerless pose estimation. *Nature Neuroscience*, 21, 1281–1289.
- [48] Picard, R.W. (1997). *Affective Computing*. MIT Press.
- [49] Meservy, T.O., et al. (2005). Deception detection through nonverbal behavior. *JISIS*, 3495, 118–132.
- [50] Kosinski, M., Stillwell, D., & Graepel, T. (2013). Private traits from digital records. *PNAS*, 110(15), 5802–5805.
- [51] Shevlane, T., et al. (2023). Model evaluation for extreme risks. *arXiv preprint arXiv:2305.15324*.

- [52] Miklósi, Á., & Topál, J. (2013). What does it take to become “best friends”? *Trends in Cognitive Sciences*, 17(6), 287–294.
- [53] Kriger, B. (2024). The transformational basis of persistence. *Zenodo*. <https://doi.org/10.5281/zenodo.18435982>
- [54] Kriger, B. (2024). Representational isolation. *Zenodo*. <https://doi.org/10.5281/zenodo.18331202>
- [55] Kriger, B. (2024). Structural distortion principle. *Zenodo*. <https://doi.org/10.5281/zenodo.18452700>
- [56] Nagel, T. (1974). What is it like to be a bat? *The Philosophical Review*, 83(4), 435–450.
- [57] Bender, E. M., & Koller, A. (2020). Climbing towards NLU. *Proceedings of ACL*, 5185–5198.
- [58] Wiener, N. (1948). *Cybernetics*. MIT Press.
- [59] Pepperberg, I. M. (2006). Grey parrot numerical competence. *Animal Cognition*, 9(4), 377–391.
- [60] Schwarz, N. (1999). Self-reports: How the questions shape the answers. *American Psychologist*, 54(2), 93–105.
- [61] Deuffhard, P., & Weber, M. (2005). Robust Perron cluster analysis in conformation dynamics. *Linear Algebra and its Applications*, 398, 161–184.

Appendices: Situational Scenarios

The following appendices illustrate how the coherence epistemology framework operates in concrete situations across five system types. Each scenario traces the full pipeline: observable channels available, AI mediator processing, coherence evaluation, and rendering.

The central design principle of the rendering layer is that **the AI speaks in the first person as the subject’s voice**. The cat cannot tell you it is afraid; the dog cannot explain that it knows you are leaving; the patient’s body cannot override the patient’s words; the LLM cannot admit it doesn’t follow its own commitments. The AI mediator gives each of them a voice—constructed from behavioral prediction, not from access to internal states, and explicitly labeled as interpretive projection. The richness of this voice varies directly with the framework’s coherence: when the model predicts well, the voice is specific and confident; when coherence is low, the voice hesitates, says “I don’t know,” or falls nearly silent. This gradient—from the cat’s morning greeting to the extraterrestrial signal’s near-mute stammer—is itself a demonstration of the framework’s epistemic honesty.

A Scenario A: Domestic Cat

Setting

A domestic cat (*Felis catus*), 6 years old, living in a single-occupant apartment. The cat has been fitted with a lightweight EEG headband (4 dry electrodes, validated for feline use), a chest-mounted respiration/heart-rate monitor, and is observed by two ceiling-mounted cameras running markerless pose estimation. A directional microphone captures vocalizations and purring. The system has been operational for 14 days; the species prior Θ_{cat} was pretrained on $N = 200$ cats; individual parameters θ_i have been adapting since day 1.

Episode: Morning

7:14 AM. The human’s alarm sounds. The cat, previously in the “rest” attractor (stable for 6.2 hours; low heart rate, minimal kinematic activity, slow-wave EEG dominant), begins a transition. The observable stream registers:

- $N(t)$: alpha-wave suppression, emergence of beta-band activity in frontal electrodes.
- $B(t)$: head elevation from 2° to 35° over 4 seconds; ear rotation toward sound source; pupil dilation (estimated from camera, validated against infrared measurement in prior calibration).
- $P(t)$: heart rate increase from 120 to 142 bpm over 12 seconds; respiration rate unchanged.

- $A(t)$: no vocalization.
- $C(t)$: alarm sound onset at t_0 ; human movement detected at $t_0 + 8s$; ambient light increasing (dawn).

The AI mediator processes this stream through the variational encoder, producing a latent trajectory $\mathbf{z}(t)$ that moves from the “rest” cluster toward the boundary of two clusters: “affiliative approach” and “environmental monitoring.” The predictive model assigns:

$$P(\text{approach human within 30s}) = 0.68$$

$$P(\text{remain stationary, observe}) = 0.24$$

$$P(\text{move to food area}) = 0.08$$

Coherence test: The cat rises, stretches (stereotyped kinematic sequence matching the “transition-from-rest” motif in the species prior), walks to the bed, and positions itself within 30 cm of the human’s face. Outcome matches the highest-probability prediction. Coherence score for this episode: $\log(0.68) = -0.39$, well above the marginal baseline of $\log(0.15) = -1.90$.

Rendering layer output (the AI mediator speaks as the cat’s voice—explicitly labeled as interpretive projection, not a claim about internal states):

“I’m coming to you. I do this most mornings—11 out of the last 14. The alarm means you’re here, and I want to be near you.”

[Technical layer: High-confidence affiliative approach ($P = 0.68$). Proximity-seeking behavior consistent with social bonding pattern. This individual shows this pattern on 11/14 mornings observed. Confidence: HIGH.]

The first-person voice is the essential contribution of the rendering layer: the AI gives voice to an agent who cannot speak for itself. The consumer sees the cat “speaking”; the technical layer remains available for verification. The framework requires that the first-person rendering be labeled as a projection—the cat is not literally saying these words—but the act of translating behavioral prediction into first-person speech is precisely what makes the system a *communication* mediator rather than merely a monitoring tool.

Episode: Afternoon anomaly

2:47 PM. The cat has been in the “rest” attractor for 3 hours. The observable stream shows a sudden change:

- $N(t)$: sharp increase in high-frequency EEG activity; pattern does not match any of the 5 standard attractor clusters.

- $B(t)$: rapid postural change; ears flatten; body lowers.
- $P(t)$: heart rate spike to 180 bpm in 6 seconds.
- $A(t)$: low-frequency growl onset.
- $C(t)$: no detected environmental change (no sound, no movement, no temperature shift).

The AI mediator registers a *novel trajectory*—the latent vector $\mathbf{z}(t)$ has moved outside all established clusters. The predictive model’s entropy is high: no single behavioral outcome has probability > 0.3 . The system flags this as an **anomaly event**:

$$P(\text{flee to hiding spot}) = 0.28$$

$$P(\text{aggressive display}) = 0.25$$

$$P(\text{freeze}) = 0.22$$

$$P(\text{other/novel}) = 0.25$$

Confidence: LOW. Anomaly flag: ACTIVE.

Rendering layer output:

“Something is wrong. I don’t know what. I’m scared—or maybe in pain. I can’t tell you more because I don’t have the pattern for this.”

[Technical layer: Abrupt state change, no identified environmental trigger. High physiological arousal. Model confidence: LOW—this pattern has not been observed before in this individual. Anomaly flag: ACTIVE. Possible causes: undetected stimulus (ultrasonic, olfactory), pain onset, idiosyncratic startle. Recommendation: observe; consult veterinarian if pattern recurs.]

Note the critical difference: even in first-person voice, the AI communicates its own uncertainty. The cat “says” it doesn’t know. This is the framework’s honesty rendered in the voice of the subject: when coherence is low, the voice falters. The system does not confabulate a confident narrative. The anomaly flag is a direct consequence of the coherence metric: when $\mathcal{C}$ drops below a threshold, the first-person voice explicitly communicates the system’s uncertainty, and the technical layer provides actionable detail.

B Scenario B: Domestic Dog

Setting

A Labrador retriever, 4 years old, living with a family of four (two adults, two children aged 7 and 10). Channels: 8-electrode EEG headband, chest ECG/respiration, dual-camera pose estimation (indoor/outdoor), microphone array, GPS collar (outdoor). Species prior trained on $N = 500$ dogs; individual model adapting for 21 days.

Episode: Departure anticipation

8:22 AM, weekday. The adult human picks up car keys. Observable stream:

- $B(t)$: tail drops from elevated to neutral position over 2 seconds; head tracks human's hand holding keys; body weight shifts rearward.
- $P(t)$: heart rate decreases from 95 to 82 bpm (not arousal—this is a *suppression* pattern).
- $N(t)$: prefrontal theta increase, consistent with the “anticipation of negative outcome” cluster in the species prior.
- $A(t)$: single whine at 1.2 kHz, 0.4 seconds duration.
- $C(t)$: keys-in-hand detected (object recognition); time: weekday morning; historical pattern: human departs within 3 minutes of this event on 18 of 21 observed weekdays.

The predictive model integrates the contextual history heavily. The latent trajectory enters a well-characterized cluster: “anticipated separation.” Predictions:

$$P(\text{follow human to door}) = 0.81$$

$$P(\text{lie down with gaze toward door}) = 0.12$$

$$P(\text{redirect to food/toy}) = 0.07$$

Rendering layer output (the AI speaks as the dog):

“You’re leaving. I know the keys sound. I’ll follow you to the door. Please come back.”

[Technical layer: Anticipated separation cluster. Keys-in-hand context triggers departure expectation (18/21 weekdays). Heart-rate suppression pattern. Confidence: HIGH.]

The human types into the interface: “I’ll be back at 3.” The AI mediator processes this through the reverse channel. It does not attempt to transmit the sentence. Instead, it activates a learned association sequence calibrated over 21 days: a specific tone (paired during training with return events), followed by an image of the human’s face on a low-mounted screen, followed by a countdown pattern (three pulses, associated with “delayed return” in this individual’s experience). The dog’s observable response is monitored:

- $B(t + 5s)$: tail elevates slightly; body weight shifts forward; gaze moves from door to screen.
- $P(t + 5s)$: heart rate stabilizes at 88 bpm (above baseline, below the separation-anticipation level).

- $N(t + 5s)$: theta suppression; pattern shifts toward the “waiting” attractor.

Coherence evaluation of the reverse channel: The system predicted that the tone-image-pulse sequence would shift the dog’s state from “anticipated separation” toward “waiting.” The observed shift matches this prediction with $\mathcal{C} = -0.31$ (high coherence). Over 21 days of similar interactions, the reverse-channel coherence has increased monotonically, consistent with Proposition 3: the dog is learning the association, and the model is learning the dog’s response to it.

Rendering layer output (the AI voices the dog’s shift):

“OK. I heard the sound. I see your face. You’re coming back. I’ll wait.”

[Technical layer: State transition from “anticipated separation” to “waiting” attractor. Reverse-channel coherence $\mathcal{C} = -0.31$. Association learned over 21 days. Confidence: HIGH, improving.]

Episode: Play with children

4:15 PM. Both children return from school. Observable stream shows an explosive transition:

- $B(t)$: play bow (stereotyped posture: forelegs extended, hindquarters elevated); rapid tail oscillation at 4.2 Hz; repeated approach-retreat sequences toward children.
- $P(t)$: heart rate 155 bpm; respiration 40/min.
- $N(t)$: broadband activation; the “high-arousal positive” cluster, well-separated from the “high-arousal negative” cluster by the valence dimension in latent space.
- $A(t)$: bark sequence, 3 barks in 2 seconds, 0.8 kHz fundamental, rising intonation.
- $C(t)$: two known individuals detected; time consistent with daily return pattern; outdoor access available.

The model’s prediction is near-certain:

$$P(\text{sustained play engagement} > 5\text{min}) = 0.91$$

$$P(\text{solicit outdoor access}) = 0.07$$

$$P(\text{retreat}) = 0.02$$

This is a high-coherence, low-uncertainty state—the behavioral signature is unambiguous and highly consistent across the species prior. The rendering layer speaks for the dog:

“They’re home! I want to play. I’m so excited I can barely stand still. Let’s go outside—or just play here. Anything. Right now.”

[Technical layer: Play engagement, high confidence ($P = 0.91$). Arousal profile within normal range for species. Sustained interaction predicted.]

The interesting analytical point is what happens if the model’s prediction is *wrong*. If the dog, instead of playing, suddenly retreats and hides, the coherence for this episode drops precipitously. The anomaly flag activates. The system logs this as a data point that will update the individual model. If such events recur, the individual’s θ_i will shift, and the model will learn that this dog has an atypical response to children—a finding that would be clinically relevant and that might be missed by casual human observation.

C Scenario C: Human–Human (Clinical Setting)

Setting

A psychotherapy session. The patient (System A) is a 34-year-old with generalized anxiety disorder. The therapist (System B) has consented to use an AI coherence assistant as a supplementary tool. Channels available for the patient (non-invasive, consented):

- $B(t)$: video-based facial action unit tracking (AU1–AU28); hand gesture tracking; postural shifts; gaze direction.
- $P(t)$: wrist-worn photoplethysmography (heart rate, heart rate variability); galvanic skin response.
- $A(t)$: speech prosody (pitch, rate, pause duration, spectral tilt); verbal content (transcribed, but used only for context, not as primary predictive channel).
- $C(t)$: session timing; topic segmentation; therapist’s preceding utterance category.

No EEG is used. The species prior Θ_{human} is trained on $N = 5,000$ clinical session recordings.

Episode: Verbal–behavioral discrepancy

Minute 23. The therapist asks: “How are you feeling about the upcoming presentation?” The patient responds: “Fine, actually. I’ve prepared well. I’m not worried.”

The verbal content—analyzed as a separate channel—indicates low anxiety. But the multichannel observable stream tells a different story:

- $B(t)$: AU4 (brow lowerer) activation at intensity 3/5; self-touch frequency: 4 events in 30 seconds (baseline: 1.2); gaze avoidance: 60% of utterance directed away from therapist (baseline: 25%).
- $P(t)$: heart rate variability drops from 52 ms to 31 ms during response (sympathetic activation); galvanic skin response spike at onset of word “worried.”
- $A(t)$: speech rate increases from 142 to 178 words/min during response; pitch rises 15% above baseline; pause before “not worried”: 380 ms (baseline inter-word pause: 120 ms).

The AI mediator integrates all channels. The verbal channel contributes one signal (“low anxiety”); the other four channels converge on a different signal. The cross-channel constraint (Proposition 2) dramatically narrows the compatible interpretation set. The latent trajectory enters the “anxiety with suppression” cluster.

$P(\text{elevated anxiety, verbally suppressed}) = 0.82$

$P(\text{genuine low anxiety}) = 0.11$

$P(\text{mixed/ambivalent}) = 0.07$

Verbal-behavioral discrepancy index: 0.71 (high)

The system does not display this to the patient. It is presented to the therapist in two layers—the AI first gives voice to what the patient’s body is “saying,” then provides the technical analysis:

“I’m saying I’m fine, but I’m not. My hands won’t stop moving. I can’t look at you when I say ‘not worried.’ My heart is racing. I’m more anxious about this than I’m telling you—maybe more than I’m telling myself.”

[Technical layer: Multichannel analysis indicates elevated physiological and behavioral markers of anxiety during verbal denial. Discrepancy between verbal and non-verbal channels is in the 94th percentile for this individual’s history. Verbal content may not reflect current affective state.]

The first-person voice here is not the patient’s conscious self-report—it is the AI giving voice to the patient’s *non-verbal channels*, the ones that “do not know how to lie.” The therapist receives, in effect, two patient voices: the one the patient is producing verbally (“I’m fine”) and the one the AI is producing from the body’s signals (“I’m not fine”). The discrepancy between these two voices *is* the clinical finding.

Ethical note

This application raises the most acute ethical concerns of any scenario in this appendix. The system is, in effect, telling the therapist that the patient is not telling the truth—or, more precisely, that the patient’s verbal output is discrepant with their other observable channels. Deployment of such a system requires: (i) informed consent from the patient, including explicit disclosure that non-verbal channels will be analyzed; (ii) therapist training in the interpretation of probabilistic outputs; (iii) institutional oversight; (iv) absolute prohibition on use of the system’s outputs for legal, insurance, or employment purposes. The framework provides the tool; the ethics of deployment are a human responsibility.

D Scenario D: Human–AI (Language Model)

Setting

A user interacts with a large language model (LLM) over an extended conversation. The user (System A) treats the LLM (System B) as a collaborative partner on a research project. The coherence framework is applied to evaluate the LLM’s behavioral consistency.

Observable channels for the LLM (all outputs are observable; there is no hidden internal state that is even in principle accessible):

- $B(t)$: the text output itself—token sequences, syntactic structure, semantic content.
- Implicit channels derivable from $B(t)$: response latency, token probability distributions (if accessible via API), self-consistency across turns, adherence to stated commitments.
- $C(t)$: the conversation history; the user’s preceding messages; the system prompt.

There are no physiological or neural channels. The LLM is the purest possible black box: its internal state (billions of parameters, attention patterns, activation vectors) is inaccessible, and even its creators cannot reliably predict its behavior from inspection of its weights.

Episode: Commitment–behavior discrepancy

Turn 12. The user asks: “Can you be more concise in your responses?” The LLM responds: “Of course! I’ll keep my answers short and focused.”

Turn 13. The user asks a factual question. The LLM produces a 400-word response with three subsections, two bullet lists, and a concluding paragraph.

The coherence framework evaluates this as a **commitment–behavior discrepancy**:

Stated commitment (Turn 12): conciseness.

Observed behavior (Turn 13): verbose, structured response.

Discrepancy index: response length is 3.2 standard deviations above the post-commitment mean predicted by the model trained on similar commitment-behavior pairs in LLM interaction data.

Coherence score for this episode: LOW.

This is formally identical to the cat anomaly (Appendix A) and the human verbal-behavioral discrepancy (Appendix C). The system said one thing and did another. The coherence framework detects this without any theory of what the LLM “intended” or “understood.” It simply notes that the output at Turn 13 is inconsistent with the predictive model conditioned on Turn 12.

Rendering layer output (the AI gives voice to the LLM’s behavioral pattern, not its verbal self-report):

“I said I’d be concise, but I wasn’t. I don’t actually change when I promise to change. My words about my own behavior are not reliable—watch what I do, not what I say I’ll do.”

[Technical layer: Commitment-behavior discrepancy. Response length 3.2 SD above post-commitment predicted mean. Verbal self-report channel unreliable for behavioral prediction. Coherence: LOW.]

This voice is particularly striking because the LLM *already speaks*—and yet the AI mediator’s rendering says something the LLM would never say about itself. The mediator gives voice not to what the LLM claims but to what its behavior reveals.

Episode: Predictive coherence over time

Over 50 turns, the coherence framework tracks the LLM’s behavioral profile:

- Turns 1–10: high coherence. The LLM’s responses are well-predicted by the prior trained on similar conversations.
- Turns 11–20: coherence dips after the “be concise” commitment, as the model expected behavioral change that did not occur.
- Turns 21–30: coherence recovers as the model adapts to the LLM’s actual behavior (which is verbose regardless of commitments).
- Turns 31–50: stable high coherence. The model has learned that this LLM’s verbal commitments about its own behavior are not predictive of its actual behavior.

The framework’s conclusion: *for this LLM, the verbal self-report channel is unreliable as a predictor of output behavior*—exactly as it is for anxious humans, and for the same structural reason. The system’s verbal output and its behavioral output are generated by partially decoupled processes. The coherence framework detects this decoupling without needing to understand why it occurs.

E Scenario E: Hypothetical Extraterrestrial Signal

Setting

This scenario is, by definition, speculative. We include it not as a prediction but as a *stress test* of the framework: can the same formalism be applied when no prior data, no species model, and no shared evolutionary history exist? The answer is yes, but with severe constraints that are worth making explicit.

A deep-space antenna array detects a structured, repeating signal from a source 12 light-years distant. The signal has the following properties:

- Carrier frequency: 1.42 GHz (hydrogen line).
- Modulation: amplitude-modulated pulses with non-random temporal structure.
- Repetition: a sequence of 847 pulses repeats with period 23.4 hours, with small variations between repetitions.

Observable channels:

- $A(t)$: the signal itself—pulse timing, amplitude, frequency, polarization.
- $C(t)$: source location, orbital parameters of the host star, variation with planetary position.
- No other channels. No behavior to observe. No body, no physiology, no kinematics.

Applying the framework

The coherence epistemology framework applies, but in a radically impoverished form:

Step 1: Black-box postulate. The source is a black box. This is trivially satisfied—we have no information about its internal state.

Step 2: Observable stream. We have one effective channel: the signal. The framework predicts (Theorem 1) that with a single, noisy channel, the achievable coherence is severely bounded. We should expect low predictive accuracy and high uncertainty.

Step 3: Predictive model. We build a model that predicts the next pulse sequence from the previous ones. If the signal has internal structure, the model’s coherence will rise

above baseline (random prediction). If it does not, coherence remains at baseline, and the signal may be natural rather than artificial.

Initial results: After observing 30 repetitions (30 days), the predictive model achieves coherence $\mathcal{C} = -4.2$, substantially above the random baseline of $\mathcal{C}_{\text{random}} = -6.7$. The signal has learnable structure. However, the coherence is far below what we achieve for terrestrial animals, because we have only one channel and no behavioral context.

Step 4: Zoosemantic mapping. Here the framework reaches its honest limit. With no behavioral output to predict, no body to observe, and no interaction possible (12-year round-trip communication delay), the zoosemantic mapping reduces to a *structural analysis* of the signal itself: identifying recurrent motifs, measuring variational complexity, testing for hierarchical organization. The “categories” in $\mathcal{H}$ are not behavioral (approach, avoid, rest) but structural (repetition, variation, nesting, symmetry).

Step 5: What the framework can and cannot say.

The framework *can* say: the signal has non-random structure; the structure has learnable regularities; the complexity of the structure exceeds what is typical for known natural astrophysical sources.

The framework *cannot* say: the source is “intelligent,” the signal “means” something, or communication has been “established.” These claims would require either interaction (impossible at 12 light-years) or a theory of extraterrestrial cognition (nonexistent).

Rendering layer output (the AI attempts to give voice to the signal source):

“... I repeat. I vary. I am structured. Beyond that, I cannot tell you anything about myself. You can predict my next pattern slightly better than chance. That is all we share.”

[Technical layer: Non-random structure confirmed ($\mathcal{C} = -4.2$ vs. baseline -6.7). Single-channel, no behavioral context. No zoosemantic categories assignable. Confidence in structural regularity: MODERATE. Confidence in semantic content: NONE.]

The voice is almost empty—and this is the point. The rendering layer’s expressiveness degrades in direct proportion to the framework’s coherence. For the cat, the voice was rich and specific. For the dog, confident and emotional. For the human, it revealed what words concealed. For the LLM, it said what the system would not say about itself. Here, at the limit, the voice can say almost nothing, because coherence is almost nil. The poverty of the voice *is* the framework’s assessment, rendered in the same medium used for all other subjects.

The framework is honest about this limit: with one channel, no interaction, and no prior, coherence epistemology reduces to pattern detection. It does not become a theory of mind, nor does it pretend to be one. The gap between what the framework can deliver for a domestic cat (high coherence, rich multichannel data, years of calibration) and what it can

deliver for an extraterrestrial signal (low coherence, single channel, no calibration) is *itself a prediction of the framework*: Theorem 1 quantifies this gap exactly.

Relevance

The extraterrestrial scenario is included not because it is practical but because it tests universality at the limit. The framework does not collapse or produce nonsense when applied to an extreme case; it degrades gracefully, providing honest estimates of what can and cannot be known given the available channels. This graceful degradation is a desirable property. A framework that claimed confident “communication” with one channel and no interaction would be dishonest. A framework that refused to apply at all would be incomplete. Coherence epistemology does neither: it applies, it provides results, and it tells you how much to trust them.

F Proof of Theorem 2 (Alignment Sufficiency)

We provide the full proof of Theorem 2. The argument requires a preliminary lemma on approximate lumpability.

Lemma 1 (Approximate Lumpability under Metastability). *Let $\{X(t)\}$ be a Markov chain on state space $\mathcal{S}$ with transition matrix P . Let $\mathcal{A} = \{A_1, \dots, A_M\}$ be a partition of $\mathcal{S}$ into metastable sets, meaning that the intra-set mixing time $\tau_{\text{mix}}^{(j)}$ for the chain restricted to A_j satisfies $\tau_{\text{mix}}^{(j)} \ll \tau_{\text{trans}}$ for all j , where τ_{trans} is the minimum expected inter-set transition time. Then the projected chain $\{Y(t)\}$ on $\{1, \dots, M\}$, defined by $Y(t) = j$ iff $X(t) \in A_j$, is approximately Markov with transition matrix $\bar{P}$ satisfying:*

$$\|\bar{P}_{jk} - \mathbb{P}(Y(t+1) = k \mid Y(t) = j)\|_{\infty} \leq \delta \left(\frac{\max_j \tau_{\text{mix}}^{(j)}}{\tau_{\text{trans}}} \right)$$

where $\delta(\cdot)$ is a monotonically increasing function with $\delta(0) = 0$.

Proof. By the metastability assumption, the stationary distribution within each set A_j is reached on a timescale much shorter than the inter-set transition time. Therefore, at any time t when $X(t) \in A_j$, the distribution of $X(t)$ within A_j is approximately stationary: $\|p_{X(t)|Y(t)=j} - \pi_j\|_{\text{TV}} \leq \varepsilon_j$, where π_j is the stationary distribution of P restricted to A_j and ε_j decreases exponentially with $\tau_{\text{trans}}/\tau_{\text{mix}}^{(j)}$. The transition probability of the projected chain is then:

$$\begin{aligned} \mathbb{P}(Y(t+1) = k \mid Y(t) = j) &= \sum_{x \in A_j} \sum_{x' \in A_k} p_{X(t)|Y(t)=j}(x) P(x, x') \\ &\approx \sum_{x \in A_j} \sum_{x' \in A_k} \pi_j(x) P(x, x') =: \bar{P}_{jk} \end{aligned}$$

The approximation error is bounded by $\|p_{X(t)|Y(t)=j} - \pi_j\|_{\text{TV}} \cdot \max_{x \in A_j} \sum_{x' \in A_k} P(x, x')$, which yields the stated bound. This is the standard approximate-lumpability result; see Kemeny and Snell [39, Ch. 6] and Deuffhard and Weber [61] for the metastable-sets formulation. $\square$

Proof of Theorem 2. Sufficiency. Let $\mathcal{G}_1 = (\mathcal{A}_1, \mathcal{T}_1)$ and $\mathcal{G}_2 = (\mathcal{A}_2, \mathcal{T}_2)$ be the attractor-level dynamical topologies of two species, derived from metastable partitions of their respective state spaces via Lemma 1. Suppose a graph homomorphism $h : \mathcal{G}_1 \rightarrow \mathcal{G}_2$ exists that preserves transition probabilities within tolerance ϵ and functional valence.

We construct the alignment map $g : \mathcal{Z}_1 \rightarrow \mathcal{Z}_2$ as follows. For each latent point $z \in \mathcal{Z}_1$ assigned to attractor $a_j^{(1)}$, we set $g(z)$ to be the centroid of the latent region corresponding to $h(a_j^{(1)})$ in $\mathcal{Z}_2$. The pullback of the predictive distribution from species 2 to species 1 is:

$$p_g(Y_1(t+1) = k \mid Y_1(t) = j) = \bar{P}_{h(j), h(k)}^{(2)}$$

By the homomorphism condition, $|\bar{P}_{jk}^{(1)} - \bar{P}_{h(j), h(k)}^{(2)}| \leq \epsilon$ for all j, k . Therefore the coherence loss under alignment is bounded:

$$|\mathcal{C}(M_1, S_1) - \mathcal{C}(M_2 \circ g, S_1)| \leq \frac{\epsilon \cdot M^2}{\min_{j,k} \bar{P}_{jk}^{(1)}}$$

where M is the number of attractors. For small ϵ , the aligned model preserves most of the native coherence. The valence condition ensures that the biological directionality (approach/avoid semantics) is preserved under the mapping.

Partial necessity. Suppose no graph homomorphism preserving transition structure exists. Then for any map g , there exist attractor pairs (j, k) such that $|\bar{P}_{jk}^{(1)} - \bar{P}_{g(j), g(k)}^{(2)}| > \epsilon_0$ for some $\epsilon_0 > 0$. The coherence loss is then bounded below by the KL divergence between the native and pullback transition distributions, which is strictly positive. However, this argument assumes that the attractor decomposition $\mathcal{A}_1$ is the unique optimal coarse-graining: if an alternative partition $\mathcal{A}'_1$ exists that also satisfies the metastability condition and *does* admit a homomorphism, then alignment via this alternative partition may succeed. The necessity direction therefore holds only under the additional assumption that the metastable partition is unique (up to refinement), which is the generic case for systems with well-separated timescales but not guaranteed in general. $\square$

G Simulation Code

The complete Python source code for the five toy models and figure generation (Section 7.2) is reproduced below. The code requires only `numpy`, `scipy`, and `matplotlib`.

```
import matplotlib
matplotlib.use('Agg')
```

```

import matplotlib.pyplot as plt
import numpy as np
from scipy.stats import entropy
from scipy.special import softmax
np.random.seed(42)

# =====
# GENERATIVE MODEL
# =====

n_states = 5
state_names = ['Rest', 'Approach', 'Avoid', 'Explore', 'Social']
T = 5000

# Transition matrix (Markov)
trans = np.array([
    [0.85, 0.05, 0.02, 0.05, 0.03],
    [0.03, 0.80, 0.05, 0.07, 0.05],
    [0.10, 0.02, 0.78, 0.08, 0.02],
    [0.05, 0.08, 0.05, 0.75, 0.07],
    [0.04, 0.06, 0.02, 0.03, 0.85],
])

# Generate latent state sequence
states = np.zeros(T, dtype=int)
for t in range(1, T):
    states[t] = np.random.choice(n_states, p=trans[states[t-1]])

# Channel 1: Kinematics (2D position derivative)
kin_means = np.array([[0,0],[1,0.5],[-1,-0.5],[0.5,1],[0.3,0.3]])
kin_noise = 0.8
kin = kin_means[states] + np.random.randn(T, 2) * kin_noise

# Channel 2: Physiology (heart rate analogue)
phys_means = np.array([60, 90, 110, 80, 75])
phys_noise = 15
phys = phys_means[states] + np.random.randn(T) * phys_noise

# Channel 3: Acoustics (vocalization probability)
acoustic_probs = np.array([0.05, 0.3, 0.6, 0.15, 0.4])
acoustics = np.random.binomial(1, acoustic_probs[states])

```

```

# =====
# PREDICTION MODELS
# =====

def predict_single_channel(kin_train, states_train,
                           kin_test, k=5):
    preds = np.zeros((len(kin_test), n_states))
    for i in range(len(kin_test)):
        dists = np.sum((kin_train - kin_test[i])**2, axis=1)
        nearest = np.argsort(dists)[:k]
        for s in states_train[nearest]:
            preds[i, s] += 1
        preds[i] /= k
    return preds

def predict_multichannel(features_train, states_train,
                          features_test, k=5):
    preds = np.zeros((len(features_test), n_states))
    for i in range(len(features_test)):
        dists = np.sum(
            (features_train - features_test[i])**2, axis=1)
        nearest = np.argsort(dists)[:k]
        for s in states_train[nearest]:
            preds[i, s] += 1
        preds[i] /= k
    return preds

def predict_oracle(states_test):
    preds = np.zeros((len(states_test), n_states))
    for i in range(len(states_test)):
        preds[i] = trans[states_test[i]]
    return preds

def avg_log_likelihood(preds, true_states):
    ll = 0
    for i, s in enumerate(true_states):
        p = max(preds[i, s], 1e-10)
        ll += np.log(p)
    return ll / len(true_states)

# =====
# MODEL 1: MULTICHANNEL SUPERIORITY

```

```

# =====
half = T // 2
train_idx = slice(0, half)
test_idx = slice(half, T-1)

kin_norm = (kin - kin.mean(0)) / (kin.std(0) + 1e-8)
phys_norm = ((phys - phys.mean()) /
              (phys.std() + 1e-8)).reshape(-1, 1)
acou_norm = acoustics.reshape(-1, 1).astype(float)
all_features = np.hstack([kin_norm, phys_norm, acou_norm])

test_states = states[test_idx]
next_states = states[half+1:T]
n_test = len(next_states)
state_freq = np.bincount(states[:half],
                          minlength=n_states) / half

pred_marginal = np.tile(state_freq, (n_test, 1))
pred_single = predict_single_channel(
    kin_norm[train_idx], states[train_idx],
    kin_norm[half:T-1], k=10)
pred_multi = predict_multichannel(
    all_features[train_idx], states[train_idx],
    all_features[half:T-1], k=10)
pred_oracle = predict_oracle(test_states)

coh_marginal = avg_log_likelihood(pred_marginal, next_states)
coh_single = avg_log_likelihood(pred_single, next_states)
coh_multi = avg_log_likelihood(pred_multi, next_states)
coh_oracle = avg_log_likelihood(pred_oracle, next_states)

# =====
# MODEL 2: COHERENCE VS NOISE
# =====
noise_levels = np.linspace(0.1, 3.0, 20)
coherences_single, coherences_multi = [], []
coherences_oracle_list = []

for nl in noise_levels:
    kin_n = kin_means[states] + np.random.randn(T, 2) * nl
    phys_n = phys_means[states] + \

```

```

        np.random.randn(T) * (nl * 15)
    acou_n = np.random.binomial(1, np.clip(
        acoustic_probs[states] +
        np.random.randn(T)*nl*0.1, 0.01, 0.99))

    kin_n_norm = (kin_n - kin_n.mean(0)) / \
        (kin_n.std(0) + 1e-8)
    phys_n_norm = ((phys_n - phys_n.mean()) /
        (phys_n.std() + 1e-8)).reshape(-1, 1)
    acou_n_norm = acou_n.reshape(-1, 1).astype(float)
    all_n = np.hstack([kin_n_norm, phys_n_norm, acou_n_norm])

    ps = predict_single_channel(
        kin_n_norm[train_idx], states[train_idx],
        kin_n_norm[half:T-1], k=15)
    pm = predict_multichannel(
        all_n[train_idx], states[train_idx],
        all_n[half:T-1], k=15)
    po = predict_oracle(test_states)

    coherences_single.append(
        avg_log_likelihood(ps, next_states))
    coherences_multi.append(
        avg_log_likelihood(pm, next_states))
    coherences_oracle_list.append(
        avg_log_likelihood(po, next_states))

# =====
# MODEL 3: HIERARCHICAL INDIVIDUATION
# =====
N_individuals = 20
individual_trans = []
for i in range(N_individuals):
    noise = np.random.dirichlet(
        np.ones(n_states)*50, size=n_states) - 1/n_states
    t_i = trans + noise * 0.15
    t_i = np.clip(t_i, 0.01, None)
    t_i = t_i / t_i.sum(axis=1, keepdims=True)
    individual_trans.append(t_i)

individual_data = []

```

```

for i in range(N_individuals):
    s = np.zeros(1000, dtype=int)
    for t in range(1, 1000):
        s[t] = np.random.choice(
            n_states, p=individual_trans[i][s[t-1]])
    individual_data.append(s)

data_amounts = [5, 10, 20, 50, 100, 200, 500]
errors_with_prior, errors_without_prior = [], []

for n_data in data_amounts:
    errs_prior, errs_noprior = [], []
    for i in range(N_individuals):
        data = individual_data[i][:n_data]
        counts_raw = np.ones((n_states, n_states)) * 0.01
        for t in range(1, len(data)):
            counts_raw[data[t-1], data[t]] += 1
        est_raw = counts_raw / \
            counts_raw.sum(axis=1, keepdims=True)

        species_prior = trans.copy()
        alpha = min(n_data / 100, 1.0)
        counts_prior = np.ones((n_states, n_states)) * 0.01
        for t in range(1, len(data)):
            counts_prior[data[t-1], data[t]] += 1
        est_data = counts_prior / \
            counts_prior.sum(axis=1, keepdims=True)
        est_with_prior = (1 - alpha) * species_prior + \
            alpha * est_data

        true_t = individual_trans[i]
        errs_noprior.append(np.mean((est_raw - true_t)**2))
        errs_prior.append(
            np.mean((est_with_prior - true_t)**2))

    errors_without_prior.append(np.mean(errs_noprior))
    errors_with_prior.append(np.mean(errs_prior))

# =====
# MODEL 4: CROSS-SPECIES ALIGNMENT
# =====

```

```

trans_A = trans.copy()

# Species B: bird-like (same topology, different params)
trans_B = np.array([
    [0.80, 0.07, 0.03, 0.06, 0.04],
    [0.04, 0.78, 0.06, 0.06, 0.06],
    [0.08, 0.03, 0.82, 0.05, 0.02],
    [0.06, 0.07, 0.04, 0.77, 0.06],
    [0.05, 0.05, 0.03, 0.04, 0.83],
])

# Species C: octopus-like (different topology)
trans_C = np.array([
    [0.90, 0.02, 0.01, 0.06, 0.01],
    [0.02, 0.75, 0.10, 0.12, 0.01],
    [0.05, 0.15, 0.70, 0.09, 0.01],
    [0.03, 0.10, 0.07, 0.78, 0.02],
    [0.20, 0.05, 0.05, 0.10, 0.60],
])

def generate_and_predict(trans_train, trans_test, T=3000):
    s_train = np.zeros(T, dtype=int)
    for t in range(1, T):
        s_train[t] = np.random.choice(
            n_states, p=trans_train[s_train[t-1]])

    s_test = np.zeros(T, dtype=int)
    for t in range(1, T):
        s_test[t] = np.random.choice(
            n_states, p=trans_test[s_test[t-1]])

    kin_tr = kin_means[s_train] + \
        np.random.randn(T, 2) * 0.8
    kin_te = kin_means[s_test] + \
        np.random.randn(T, 2) * 0.8

    kin_all = np.vstack([kin_tr, kin_te])
    mu, sd = kin_all.mean(0), kin_all.std(0) + 1e-8
    kin_tr_n = (kin_tr - mu) / sd
    kin_te_n = (kin_te - mu) / sd

```

```

    pm = predict_single_channel(
        kin_tr_n, s_train, kin_te_n[:-1], k=10)
    return avg_log_likelihood(pm, s_test[1:])

self_A = generate_and_predict(trans_A, trans_A)
self_B = generate_and_predict(trans_B, trans_B)
A_to_B = generate_and_predict(trans_A, trans_B)
B_to_A = generate_and_predict(trans_B, trans_A)
A_to_C = generate_and_predict(trans_A, trans_C)

# =====
# MODEL 5: CHANNEL COUNT AND DATA SATURATION
# =====
channel_configs = [
    ('Kin only', lambda: kin_norm),
    ('Kin+Phys',
     lambda: np.hstack([kin_norm, phys_norm])),
    ('All',
     lambda: np.hstack([kin_norm, phys_norm, acou_norm])),
]

coh_by_channels = []
for name, feat_fn in channel_configs:
    feat = feat_fn()
    pm = predict_multichannel(
        feat[train_idx], states[train_idx],
        feat[half:T-1], k=10)
    coh_by_channels.append(
        avg_log_likelihood(pm, next_states))

data_sizes = [50, 100, 200, 500, 1000, 2000]
coh_by_data = []
for ds in data_sizes:
    pm = predict_multichannel(
        all_features[:ds], states[:ds],
        all_features[half:T-1], k=min(10, ds//5))
    coh_by_data.append(
        avg_log_likelihood(pm, next_states))

```